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Impact of Climate Change on Civil Infrastructure

Master's Thesis

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Abstract

Mulik, N.V. Impact of Climate Change on Civil Infrastructure: Master's Thesis / Mulik, N.V.; Jansons, L. (2026) Riga: RTU FEEM professional master's study program "Civil Construction and Real Estate Management", 78 p.

The master's thesis consists of an introduction, three main parts, a conclusion, and proposals. The volume of the thesis is 78 pages, including 16 figures, 11 tables, 12 formulas, and a bibliography with 5 sources in Latvian and 46 sources in the English language; the paper has no appendices.

The research aims to critically review existing academic evidence on the impact of climate change on civil infrastructure, as well as to examine adaptation measures and governance approaches that could support the socio-economic adaptation to the impacts of climate change.

Part 1 of the Master's thesis consists of an analysis of available scientific information on key climate hazards and their documented effects on civil infrastructure within transport networks, urban drainage, structural materials, underground infrastructure, and resilience governance frameworks; a comparison of vulnerability rankings of the Baltic States and EU vulnerability benchmarks.

Part 2 of the Master's thesis addresses the conceptual and regulatory backbone of climate risk assessment for infrastructure, including the framework of the Intergovernmental Panel on Climate Change, Eurocodes and international standards ISO 14090/14091, methods for quantitative risk assessment, lifecycle cost assessment, adaptation strategies, the economic evaluation of adaptation strategies, nature-based solutions, digital solutions for monitoring, systemic interdependencies of infrastructure, governance challenges, and international best practices in the Netherlands, Finland and the United Kingdom.

Part 3 of the Master's thesis conducts three case studies of a quantitative nature within the infrastructure context with the help of computational models: Quantitative evaluation of flood risk to the urban drainage system of the city of Riga in the future for projections of increased intensity of rainfall, thermal expansion analysis of bridge joints throughout the road network, and assessment of coastal erosion and storm surge risk to infrastructure along the Gulf of Riga for projections of increased future temperature levels. Also included is an integrated risk prioritization matrix, an implementation and EU funding framework, and an ISO 14090-aligned monitoring and evaluation framework.

The most important conclusions and proposals resulting from the master's thesis are the following: Latvia is warming faster than the EU average (+1.4 ˚C since 1961 vs. EU +1.0 ˚C),

and all key vulnerability indicators exceed EU-27 averages. Riga's combined sewer system will face a 77 % design discharge deficit under the 2070 RCP 4.5 scenario, requiring trunk main diameters to be approximately tripled. Freeze-thaw cycling frequency has increased by 46% since 1961–1990, reducing effective pavement service life by approximately 34%. The estimated annual climate damage cost to Latvian civil infrastructure is approximately €46 million, growing at 28 % over the 2000–2022 period. Climate adaptation investments are economically justified, with benefit-cost ratios of 2.2:1 to 4.1:1. Total priority adaptation investment is approximately €53 million over 10 years, with a net national budget contribution of €8 to 14 million after EU co-financing. The primary barrier to adaptation is institutional rather than technical; governance reform, design norm revision, and cross-sectoral coordination are the most impactful near-term actions.

Keywords: climate change, civil infrastructure, infrastructure resilience, climate adaptation, Latvia, Baltic states, urban drainage, bridge engineering, coastal erosion, flood risk.

Anotācija

Mulik, N.V. Klimata pārmaiņu ietekme uz civilo infrastruktūru: maģistra darbs / Mulik, N.V.; Jansons, L. (2026) Rīga: RTU IEVF profesionālā maģistra studiju programma "Civilās būvniecības un nekustamā īpašuma pārvaldība", 78 lpp.

Maģistra darbs sastāv no ievada, trim galvenajām daļām, secinājumiem un priekšlikumiem. Darba apjoms ir 78 lappuses, ieskaitot 16 attēlus, 11 tabulas, 12 formulas un bibliogrāfiju ar 5 avotiem latviešu valodā, 46 avotiem angļu valodā; darbam nav pielikumu.

Pētījuma mērķis ir kritiski pārskatīt esošos akadēmiskos pierādījumus par klimata pārmaiņu ietekmi uz civilo infrastruktūru, kā arī izpētīt adaptācijas pasākumus un pārvaldības pieejas, kas varētu atbalstīt sociālekonomisko pielāgošanos klimata pārmaiņu ietekmei.

Maģistra darba 1. daļa sastāv no pieejamās zinātniskās informācijas analīzes par galvenajiem klimata apdraudējumiem un to dokumentēto ietekmi uz civilo infrastruktūru transporta tīklos, pilsētu kanalizācijā, strukturālajos materiālos, pazemes infrastruktūrā un noturības pārvaldības sistēmās; Baltijas valstu ievainojamības reitingu un ES ievainojamības kritēriju salīdzinājums.

Maģistra darba 2. daļa aplūko klimata riska novērtēšanas konceptuālo un regulatīvo pamatu infrastruktūrai, tostarp Klimata pārmaiņu starpvaldību paneļa ietvaru, Eirokodeksus un starptautiskos standartus ISO 14090/14091, kvantitatīvās riska novērtēšanas metodes, dzīves cikla izmaksu novērtējumu, adaptācijas stratēģijas, adaptācijas stratēģiju ekonomisko novērtējumu, uz dabu balstītus risinājumus, digitālos risinājumus uzraudzībai, infrastruktūras sistēmisko savstarpējo atkarību, pārvaldības izaicinājumus un starptautisko labāko praksi Nīderlandē, Somijā un Apvienotajā Karalistē.

Maģistra darba 3. daļa veic trīs kvantitatīva rakstura gadījumu izpētes infrastruktūras kontekstā, izmantojot skaitļošanas modeļus: kvantitatīvs plūdu riska novērtējums Rīgas pilsētas drenāžas sistēmai nākotnē, prognozējot paaugstinātu nokrišņu intensitāti, tiltu šuvju termiskās izplešanās analīze visā ceļu tīklā un piekrastes erozijas un vētras uzplūdu riska novērtējums infrastruktūrai gar Rīgas līci, prognozējot paaugstinātu temperatūras līmeni nākotnē. Iekļauta arī integrēta riska prioritāšu matrica, ieviešanas un ES finansējuma sistēma, kā arī ar ISO 14090 saskaņota uzraudzības un novērtēšanas sistēma.

Svarīgākie secinājumi un priekšlikumi, kas izriet no maģistra darba, ir šādi: Latvija sasilst straujāk nekā vidēji ES (+1,4 ˚C kopš 1961. gada pret ES +1,0 ˚C), un visi galvenie ievainojamības rādītāji pārsniedz ES-27 vidējos rādītājus. Saskaņā ar 2070. gada RCP 4.5 scenāriju, Rīgas apvienotajai kanalizācijas sistēmai būs 77 % projektētās izplūdes deficīts, kas prasīs aptuveni trīskāršot maģistrālo cauruļvadu diametrus. Saldēšanas-atkušanas ciklu biežums

kopš 1961.–1990. gada ir palielinājies par 46 %, samazinot seguma efektīvo kalpošanas laiku par aptuveni 34 %. Paredzamās ikgadējās klimata postījumu izmaksas Latvijas civilajai infrastruktūrai ir aptuveni 46 miljoni eiro, pieaugot par 28 % laikposmā no 2000. līdz 2022. gadam. Investīcijas klimata adaptācijā ir ekonomiski pamatotas, un ieguvumu un izmaksu attiecība ir no 2,2:1 līdz 4,1:1. Kopējās prioritārās adaptācijas investīcijas ir aptuveni 53 miljoni eiro 10 gadu laikā, un neto valsts budžeta ieguldījums pēc ES līdzfinansējuma ir 8 līdz 14 miljoni eiro. Galvenais adaptācijas šķērslis ir institucionāls, nevis tehnisks; pārvaldības reforma, projektēšanas normu pārskatīšana un starpnozaru koordinācija ir visietekmīgākās īstermiņa darbības.

Atslēgvārdi: klimata pārmaiņas, civilā infrastruktūra, infrastruktūras noturība, adaptācija klimata pārmaiņām, Latvija, Baltijas valstis, pilsētu drenāža, tiltu inženierija, piekrastes erozija, plūdu risks.

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Introduction

The physical infrastructure is the concrete support of contemporary society. All the roads, bridges, railways, drainage systems, tunnels, and even the buildings that encompass the cities are built on the engineering assumptions that were designed based on historical climate data. These assumptions were incorporated into design codes, maintenance plans, and asset-management records, which can endure for decades, but are now being challenged by the changing climate. Already, the global mean surface temperature of the planet has increased by around 1.1 ˚C compared to pre-industrial temperatures over the past century, and is forecast to increase by 2˚C to 4˚C by 2050 (European Environment Agency, 2017). Along with the warming happening, frequent changes in the quantity of precipitation, its duration, the increased rate of sea-level rise, and a statistically increased occurrence and intensity of extreme weather events.

This is not an issue of the future now, but we are already observing the outcomes as infrastructure breakdowns, premature degradation, and service failure. For example, in Europe, a large number of flood-related disturbances have been experienced by transport infrastructure (Pregnolato, Ford, Wilkinson, *et al.*, 2017a). The economic loss associated with road networks as it being clogged by heavy rain and river floods, further leading to delays, vehicle damage, and emergency response expenditures (Pregnolato, Ford, Glenis, *et al*., 2017b). The threat to coastal and low-lying areas increases as the sea level continues to rise, which intensifies the storm surges. Increased temperatures and growing CO_2 levels accelerate the carbonation of concrete and increase the corrosion rate of steel reinforcement (Bastidas-Arteaga, Stewart, 2015; Pedram *et al.*, 2022). Current design guidelines do not account for additional thermal and hydrological loads and pressures on subsurface infrastructure, such as tunnels, sewers, and foundations. (Spyridis, Bergmeister, 2024).

This issue is even larger as civil infrastructures are long-lasting. A bridge built today might still be in operation in the 2070s or later, whereas a motorway constructed in the 1960s was designed for climatic conditions now known to be obsolete. The fact that the rate of change of climate and the lifespan of the infrastructure lack coordination poses an acute issue of engineering and governance. The U.S. roads, bridges, coastal projects, and urban drainage infrastructure are estimated to cost more due to climate change, with a drastic increase in cost during the second half of the twenty-first century (Neumann *et al.*, 2015). EU-wide

infrastructure damage related to climate is occurring in large numbers and will continue to increase in all feasible emissions scenarios (European Environment Agency, 2017).

Nevertheless, the civil-engineering professionals and the governing structures that authorize and oversee infrastructure have not fully acted in response to this problem. Adaptation strategies exist in theories like updated design standards, nature-based solutions, and resilient materials, but are yet to be directly incorporated into daily practices (Mees, Driessen, 2019). The systems of governance are weak at both the spatial and sectoral levels, and there is a chronic lack of financing for climate adaptation in infrastructure in comparison to the anticipated damage. It is thus necessary that engineers, planners, asset managers, and policy makers have a comprehensive understanding of how climate interacts with infrastructure.

It is this context that forced to consider the impact of climate change (temperature fluctuations, heavy rain, drought, and sea level rise) and what adverse effects it exerts on the performance and resilience of the civil infrastructures, as well as exploring different adaptive design strategies and mitigation measures that can enhance resilience and sustainability. The discussion draws together the peer-reviewed literature in transport engineering, structural engineering, hydrology, and climate science with the goal of gaining a multi-disciplinary perspective of how complex this problem is.

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Aim of the Master's Thesis: to critically review and synthesize the available academic evidence on the effects of climate change on civil infrastructures and what adaptation strategies and governance mechanisms can assist in coping with these effects.

The targeted aim can be achieved by following set of tasks:

1. To list the primary physical climate risks and their effects on the civil infrastructure and survey the existing literature on their expected scale and location.
2. To examine the documented and projected effects of climate change on different types of civil infrastructures (surface and subsurface) and materials.
3. To assess the existing resistance frameworks and approaches based on technical and governance levels and identify best practices and gaps.
4. To integrate the evidence and make recommendations on the necessity of enhancing climate resilience practices for civil infrastructures.

Research object of the Master's Thesis: Civil infrastructure, which is affected by climate change.

Research subject of the Master's Thesis: Climate change, which affects the civil infrastructure.

Hypothesis of the Master's Thesis: Current civil infrastructure design standards and practices, in general, are not adequate to handle the estimated increase in the intensity of climate-related risks in the next 50 years. But the specific adaptation strategies, if provided in a systematic manner, can be effective in significantly reducing infrastructure vulnerability and lowering their life cycle expenditure. The following research questions are answered in the thesis:

1. What are the main climate changes projected for the next 50 years?
2. Is the current civil infrastructure sufficiently resilient and prepared for the climate conditions projected to occur in the next 50 years?
3. If civil infrastructure is not adequately prepared, what specific measures, upgrades, or policies should be implemented to address these shortcomings?

Methodology of the Master's Thesis: The methodology of the thesis is based on an interdisciplinary research approach focused on the evaluation of the impact of climate change on civil infrastructure. It combines qualitative literature analysis with quantitative engineering assessment methods to explore and map climate-related risks, infrastructure vulnerabilities, and adaptation strategies relevant to the Baltic region and Latvia.

The thesis begins with a literature review, including peer-reviewed scientific publications, technical reports, policy documents, and international standards related to climate change adaptation and infrastructure resilience. Academic databases such as Scopus, ScienceDirect, Google Scholar, and Web of Science were used to identify relevant sources. Additional contextual information was obtained from reports published by the Intergovernmental Panel on Climate Change (IPCC), the European Environment Agency (EEA), the European Commission, and Latvian environmental and meteorological institutions.

The theoretical framework of the thesis is based on the IPCC hazard-exposure-vulnerability risk assessment model, which was applied to evaluate climate-related infrastructure risks. Regulatory frameworks such as Eurocodes, ISO 14090 and ISO 14091 were also analyzed to assess their role in climate-resilient infrastructure planning and management.

The practical component of the thesis applies quantitative engineering methodologies through three case studies within the Latvian infrastructure context. Hydrological analysis, thermal expansion calculations, flood return period assessment, lifecycle cost analysis, and coastal erosion modelling were used to evaluate projected climate impacts under Representative Concentration Pathway (RCP) scenarios. Case Study 1 evaluates urban drainage performance in Riga under projected increases in precipitation intensity and examines adaptation options such as blue-green infrastructure and pipe system upgrades. Case Study 2 analyzes the effect of

increasing temperatures on bridge expansion joint requirements across the Latvian road network. Case Study 3 investigates coastal erosion and storm surge risks along the Gulf of Riga coastline using shoreline recession analysis and the Bruun Rule methodology.

To support decision-making, the thesis also includes cost-benefit analysis and lifecycle cost assessment methods to compare the economic efficiency of adaptation strategies over long-term infrastructure service lives. Risk prioritization matrices and resilience evaluation principles were used to identify the most critical infrastructure vulnerabilities and adaptation priorities.

Figure 1 gives an idea of the entire methodology adopted in completing the thesis, where the focus is on understanding the Data gaps and suggesting recommendations according to the findings.

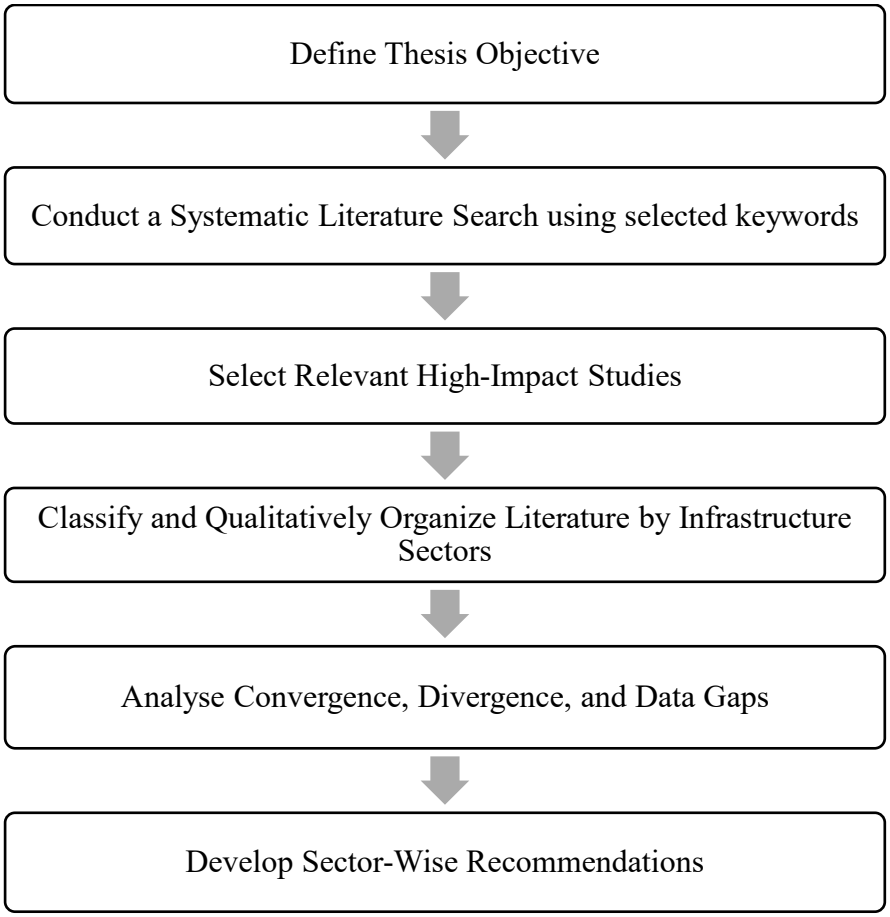


Figure 1 Methodological Flowchart (developed by author).

Scope and Limitations of the Master's Thesis: The thesis focuses on physical systems, such as transport, drainage, and engineered construction, that constitute civil infrastructure. The research does not focus exclusively on energy and telecommunications infrastructure, although both are interdependent with the categories under review. The focus is biased to Europe and

North America rather than being international, indicating the availability of literature evidence. The time Frame spans from the currently observed effects to future predictions through mid-century and the end of the century.

Resilience, or the ability of a system to withstand the disturbance, adjust, and evolve in response to stress without causing any change in serving important functions (Y. Zhang *et al.*, 2024). It is considered the organizing concept for assessing both technical and governance responses to climate-driven infrastructure stress.

Structure of the thesis: After the introduction to the topic and the aim and objectives, Part 1 is the Analytical part, which reviews the literature to identify the sectors of the infrastructure that have the greatest climatic impact. Part 2 addresses the theoretical part, examining the scientific and regulatory frameworks under transportation infrastructure and urban drainage & stormwater systems, structural durability, aiming at reinforced concrete, covering tunnels and underground infrastructure. Part 3 presents the practical part presents three case studies with full engineering, including resilience plans, governance, policies, adaptation strategies, and barriers to effectively implementing them. It also presents an integrative discussion along with conclusions and recommendations. A full reference list closes the thesis.

1. Analytical Part

Climate change is damaging the civil structures that we depend on. Heavy rains, hotter days, rising sea levels, and the increased frequency of extreme events are the results of anthropogenic warming and are affecting the long lifespan of roads, bridges, tunnels, drainage systems, and construction materials. Peer-reviewed articles and a report published by different government and non-government organizations were studied to understand how climate change is impacting infrastructure, what the existing data exposes, and how governments are planning to fight against it. Sources are categorized under six topics, namely (1) the overall picture of physical hazards and exposure, (2) effects on the transport network, (3) stormwater and urban drainage systems, (4) Material durability, (5) underground systems and tunnels, (6) resilience framework and policies.

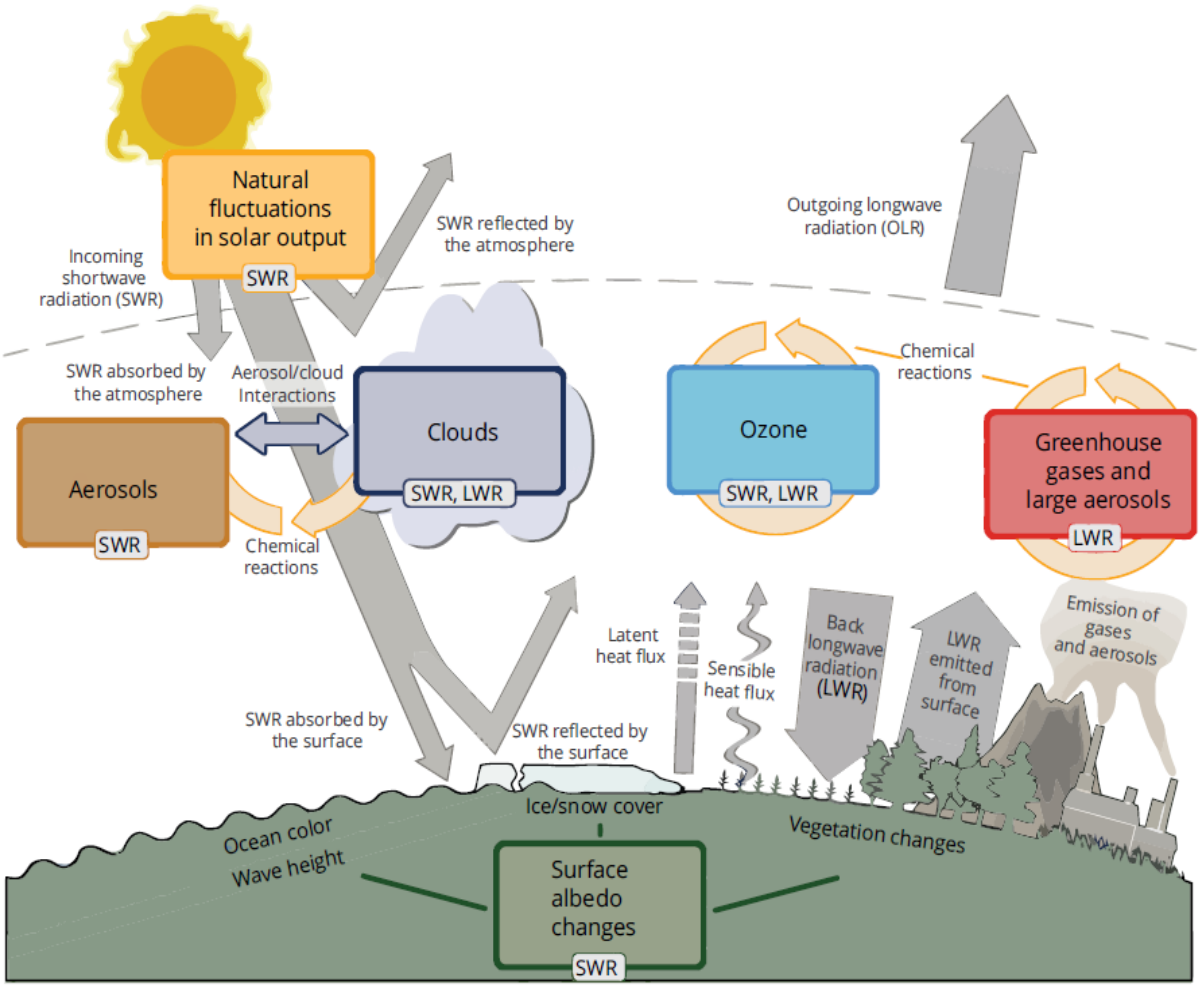


Figure 1.1 Earth's energy balance and the drivers of climate change (European Environment Agency, 2017).

The major factors affecting climate can be seen in Figure 1.1, with a comprehensive overview of the Earth's radiative energy balance. The atmosphere reflects incoming shortwave radiation (SWR) from the sun, which is partially absorbed by the aerosols, clouds, and ozone or absorbed by the surface of the Earth. The surface then radiates longwave radiation (LWR), which is again partially absorbed by greenhouse gases and clouds, and the rest escapes to space as outgoing longwave radiation (OLR). The most important processes are aerosol-cloud interactions, chemical reactions, latent and sensible heat fluxes, and surface albedo variations that are affected by ice/snow covers, ocean properties, and vegetation. The figure highlights the natural variations in solar output and the essential feedback system that stabilizes the temperature and maintains energy balance on Earth.

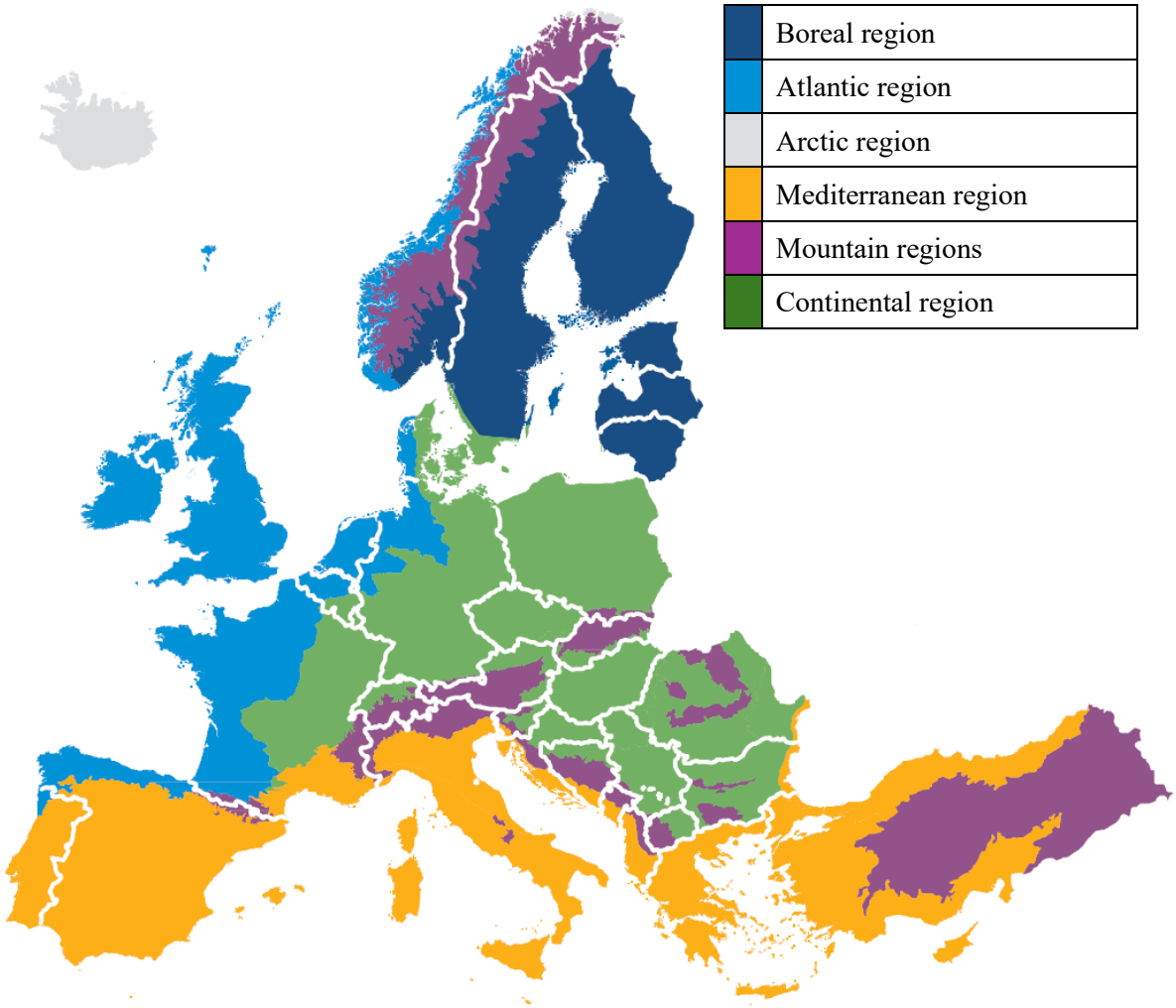


Figure 1.2 Key climate change impact and vulnerability in European regions (European Environment Agency, 2017).

The evident climate change observations and projections are shown in Figure 1.2, which divides Europe into seven main biogeographical regions. The inclusion of specific climatic changes and impacts reflects a qualitative assessment of their relative importance for the majority of a particular region (European Environment Agency, 2017).

Boreal region (Northern Europe)

Projections suggest that there will be a larger-than-average temperature increase, particularly in winter; an increase in annual precipitation and river flows; less snow; and greater damage from winter storms in this region. However, in northern Europe, climate change presents some opportunities, such as wider varieties of crops and yields, plus better growth of forests, increased generation of electrical power from water, reduced demand for energy for heating, and maybe even improved tourism opportunities in summer.

The region is predicted to be impacted by extreme weather events that could negatively affect yields, such as more variable yields, or put forests at greater risk of pest and fire attack. Heavy precipitation events are expected to become more frequent, and so will be the number of increased events.

Mountain regions

Many mountain regions are projected to experience a large increase in temperature, as well as reduced snow cover, loss of glacier mass, thawing of permafrost, and changing precipitation patterns, including less precipitation falling as snow. Mountain ecosystems are particularly vulnerable to climate change. Impacts include a change of vegetation zone and a high loss of biodiversity. Species that depend on plants and animals and live near mountain tops are in danger of extinction because they can't move to the higher elevations.

With respect to their water resources, most mountain regions are anticipated to suffer from a negative impact. The retreat of nearly all the glaciers has an impact on the water supply downstream. Other effects are less potential for winter tourism, especially in low-lying areas, higher hazard to infrastructure and settlement due to floods, landslides, and rock falls in some areas. Hydropower potential is expected to vary across the regions, benefiting some areas (like Scandinavia) and adversely affecting others.

Atlantic region (North-Western Europe)

Coastal flooding has affected low-lying coastal areas in north-western Europe in the past. This risk is likely to rise due to sea level rise and potentially more intense storm surges, with a

threat to countries of the North Sea. Greater extreme precipitation events, specifically during the winter months, are expected to lead to more river flooding, urban flooding, as well as flooding related impacts during the winter and spring months. They are forecasting that the threats of severe winter storms and perhaps severe fall storms will rise.

Continental region (Central and Eastern Europe)

Increasing heat extremes are a key hazard in central and eastern Europe. In addition to decreased summer precipitation, they can contribute to some rise in drought risk, health risk, and energy demand during the summer. The magnitude and number of river floods during the winter and spring seasons are expected to grow in different areas due to responses to projected increases in winter precipitation. Climate change will also cause an increase in the likelihood of flood events happening on rivers; crop yield variability will increase, and forest fires will become more frequent.

Arctic region (Northern Europe)

Because of the faster-than-average rise in air and sea temperatures, the Arctic is undergoing major changes that will affect both ecosystems and human activities. Loss of habitat for flora and fauna is already occurring to a limited extent. Arctic vegetation zones are likely to shift further, having wide-ranging secondary impacts. Arctic populations are suffering reductions in some species that are important to people in the North, and that are economically and ecologically significant worldwide. Acidity of marine ecosystems could be a significant problem, and the colder the water and the larger the inflow of freshwater, the more rapidly the acidification will occur in the Arctic oceans.

The Arctic is home to indigenous people who practice traditional livelihoods. These livelihoods rely directly on ecosystem services, and local communities are already affected by climate change. Traditional livelihoods, such as reindeer herding, that are under pressure from various socio-economic and political developments may suffer further from climate change impacts.

Communities are vulnerable to sea level rise and melting arctic permafrost, which can impede the economic sectors and activities, including forestry and mineral development. For the Arctic Ocean, conditions for shipping will be more favourable in the future, and there will be more opportunities to exploit non-renewable natural resources, but there are numerous risks to the environment connected to these new opportunities. Exploitation of Arctic oil and natural gas resources would be a substantial hurdle for the transition towards a low-carbon society,

given that the 2 ˚C warming limit of the United Nations Framework Convention on Climate Change (UNFCCC) implies that two-thirds of all fossil resources need to be left in the ground.

Coastal zones and regional seas

The risk of inland flooding due to rising sea levels and perhaps greater storm surges is growing throughout the coasts of Eastern Europe. Due to warming and ocean acidification, climate change is causing significant ecosystem changes in the marine environment. It can also cause a worsening of eutrophication-induced oxygen depletion, resulting in dead zones. Fisheries impacts can be positive or negative and may be greatest for the fisherman-dependent coastal fisheries that have less adaptation capacity. Higher temperatures in the water can also impact water quality (e.g., algal blooms) and can promote the spread of water-borne diseases (e.g., vibriosis).

Mediterranean region (Southern Europe)

A significant decline in precipitation and rise in temperature, especially during the summer, is observed in the Mediterranean region. The main impacts are decreases in water availability and crop yields, increasing risks of droughts and forest fires, biodiversity loss, adverse impacts on human health and well-being, and on livestock. Environmental water flows, which are important for aquatic ecosystems, are threatened by climate change and by socio-economic developments. Overall, the competition between different water users is expected to increase. Part of the reason why alien species are observed in the Mediterranean Sea and refuse to die out is the rising sea surface temperature. Climate-low water availability and high-water needs for energy generation, as well as lower water needs for energy generation, will impact the energy sector in general, but more specifically during the summer period due to water needed for cooling energy generation. Tourism suitability will decrease significantly in the peak summer season and improve in other seasons. The Mediterranean area has been identified as a vulnerable region in terms of climate change impact, with a record number of sectors facing serious impacts. It is also particularly vulnerable to the spill-over effects of climate change impacts in neighbouring regions, in particular related to disruptions in agricultural trade and to migration flows.

1.1 Climate Hazards and Infrastructure Exposure

A 2017 indicator-based report was published by the European Environment Agency, which states an average increase of 1.5˚C since pre-industrial times, and even the rain patterns

are also regionally affected as northern Europe gets slightly wetter and the southern part experiences severe storms. The sea level has increased in all the European coastal regions, with frequent storm surges. Figure 1.3 shows the sea level measurements from 1992 to 2014 across the European region observed from a satellite altimeter. Under infrastructure, the European Environment Agency (EEA) recognizes transport, building, coastal assets, and drainage as being at higher risks and confirms that the damages related to climate change are already being experienced.

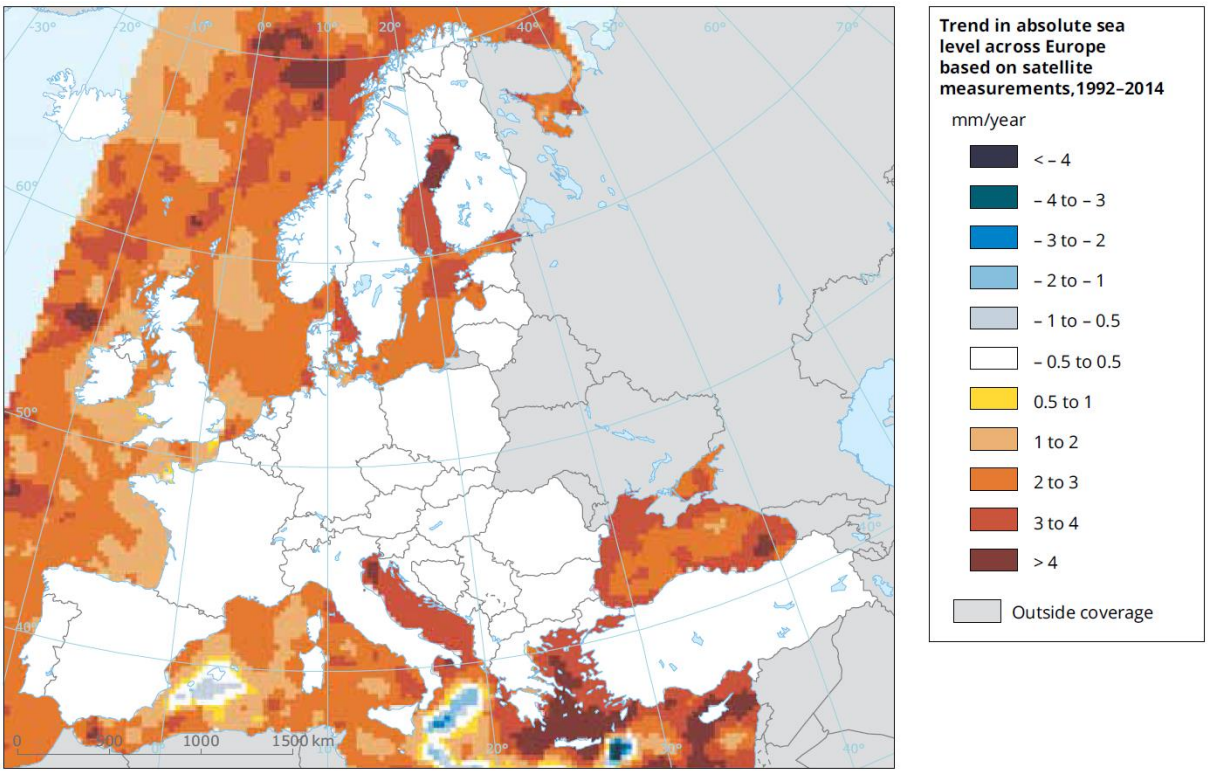


Figure 1.3 Absolute sea level across Europe (European Environment Agency, 2017).

Neumann *et al.* (2015) provided the highest number of exposures, with four large models used for estimating the toll of climate change on U.S. Roads, bridges, coastal development, and urban drainage, depending on various emission scenarios. A point to mention is that after counting the dynamic ice-melting scenario, the coastal impact can increase by 70 to 80 % compared to the models that do not count ice-melting scenarios, which means many existing readings are underestimated. On the other hand, aggressive mitigation reduces the effects by 20 to 35 %. Chinowsky, Arndt (2012) demonstrated using a stressor-response framework of roads that a developing nation is affected more by climate change than a developed nation, since they have lower adaptive capacity despite equal physical stress. Paranunzio *et al.* (2019) addressed the most neglected risk, which is the slope failure in the Italian Alps. The researchers

demonstrated that the temperature variability, including the one preceding a failure are stronger predictor of hillside instability than precipitation alone. This highlighted the geotechnical risks to the rail embankment and road cutting, other than just precipitation.

1.2 Disruption in Transport Infrastructure

Road Network Flooding

Pregnolato, Ford, Wilkinson, *et al.*, (2017a), and Pregnolato, Ford, Glenis, *et al.*, (2017b) presented the strongest studies on the relationship between flood and road, developing a function between the depth of standing water on the road and maximum safe vehicle speed, based on experience and real-world data. It describes the ability of flood to gradually decrease the capacity of the road rather than merely being open or closed. When they inserted that function into a simulation of Newcastle Upon Tyne, UK, they discovered that even a minor increase in the precipitation can result in a massive increase in network failure due to the sensitivity of some links. Weakest areas and the most appropriate location to allocate resources were identified, giving planners honest and practical inputs.

Dikanski *et al.* (2017) concentrated on bridge scouring, i.e., the erosion of the material under the foundations of the bridge during projected flood peaks. Did the monstrance that even a small increase in flood frequency can increase scour depth and foundation risk due to non-linear hydraulics, i.e., scour risk can increase more rapidly than the trend lines imply? Chinowsky *et al.* (2011) provided the most general overview of transport infrastructure, offering a four-property resilience framework – robustness, redundancy, resourcefulness, and rapidity - to all road, rail, bridge, and port sectors, suggesting assessing and managing resilience at the network level rather than the individual asset level. A Metric that reflects the effect of climate change on the country, while not overburdening the total cost, was established, along with an opportunity cost associated with the country. Opportunity cost is defined by Formula (1.1) (Chinowsky *et al.*, 2011):

$$OC_x = \frac{CC_x / SRC_x}{PR_x} \tag{1.1}$$

Where *X* is a specific country, *OC* is the opportunity cost percentage, *CC* is the total estimated climate change cost, including both maintenance and new cost, *SRC* is the cost of constructing a kilometer of new secondary paved road, and *PR* is the current paved road inventory within our country in kilometers. The formula shows that the opportunity cost of a

country is the sum of all the percentage growth in the paved road network that the country could otherwise have attained if the funds were not redirected to climate change adaptation. (Chinowsky *et al.*, 2011)

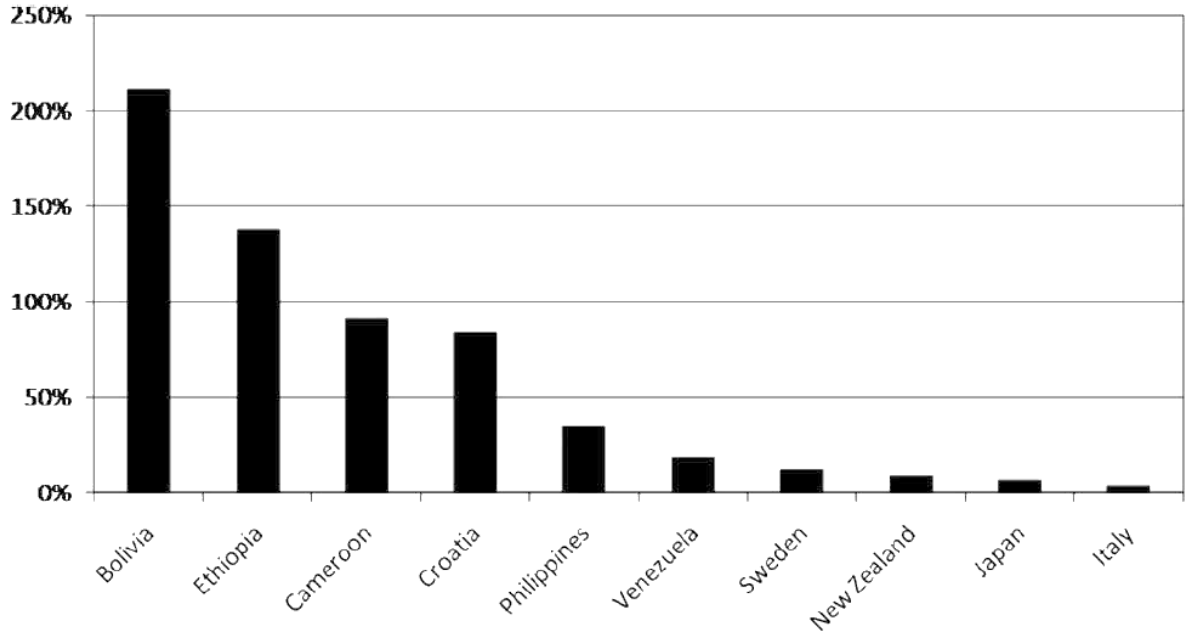


Figure 1.4 The cumulative opportunity costs as a percentage of opportunity lost to expand the existing paved road network affected by climate change in lower-income countries versus higher-income countries (Chinowsky *et al.*, 2011).

Figure 1.4 shows that climate change has significantly affected the developing countries, using the opportunity cost as the main scale of impact. Through economic and infrastructure statistics, the author brings out the fact that the three worst-hit nations are all low-income countries, meaning that they incur costs, which are at least ten times greater than those of those at the lowest end of the pyramid, which are related to climate. This is explained by the fact that these countries are already underperforming in terms of GDP and the quality of infrastructure, including paved road networks and expenditure on their maintenance, which means that they have fewer resources to mitigate environmental shocks. Even having Croatia, an upper-middle-income nation, in the top five is portrayed as an exception that proves the rule since it is an economically weak country in comparison to the other immediate European countries. (Chinowsky *et al.*, 2011)

Adaptive Transport Design

Two systematic reviews complete the transport cluster. Zhang *et al.* (2022) verified that the key institutional gap in the existing practice is the inclusion of climate information in the entire infrastructure lifecycle. Zhou *et al.* (2013) recorded climate-adaptive design approaches

for transportation infrastructure and demonstrated that the initial higher cost of adapting specifications is generally compensated through reduced maintenance and extended service life. Both reviews concluded that the obstacles to adapting practices for climate change in transportation infrastructure are primarily institutional rather than technical.

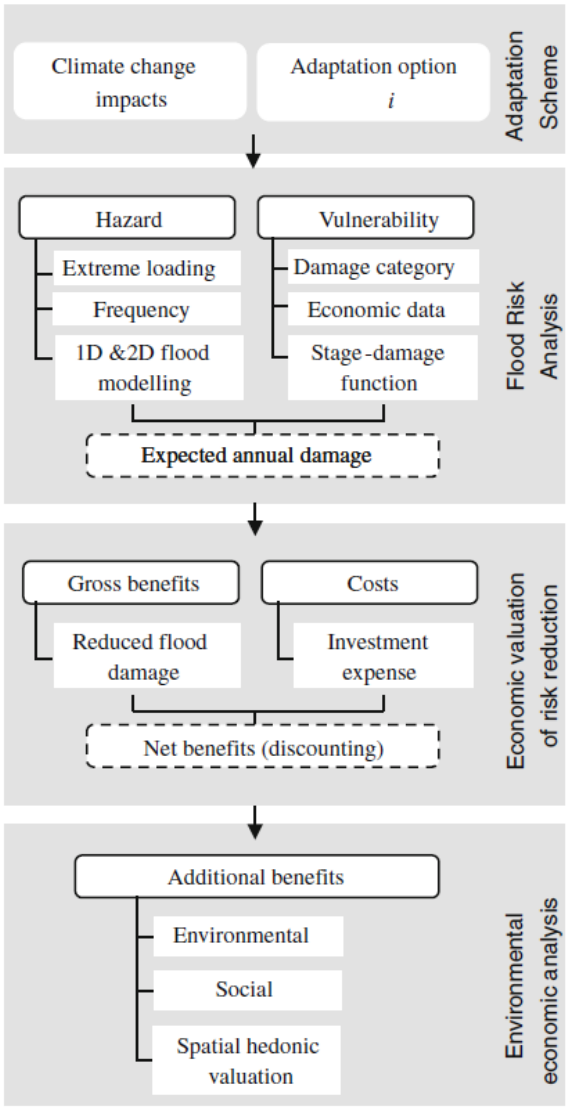


Figure 1.5 The stepwise procedure of the cross-disciplinary framework for evaluating the alternative adaptation strategies (Zhou *et al.*, 2013).

The general Procedure of the cross-disciplinary framework is shown in Figure 1.5. It includes a complete urban inundation model and multiple detailed economic models. The adaptation scheme explains the expected change of climate in an area as well as the planned adaptation alternatives. Analysis of flood risk is prepared using a flood risk assessment framework based on the estimations of both hazard and vulnerability features of the location in the setup of the adaptation strategies investigated. Risk reduction is economically valued by applying a step-by-step method to sum up the gross benefit and costs of the strategy of adjusting

to the risk reduction. Figure 1.6 shows the correlation between flood return period and the percentage of road damage for both road types, i.e., paved and unpaved roads.

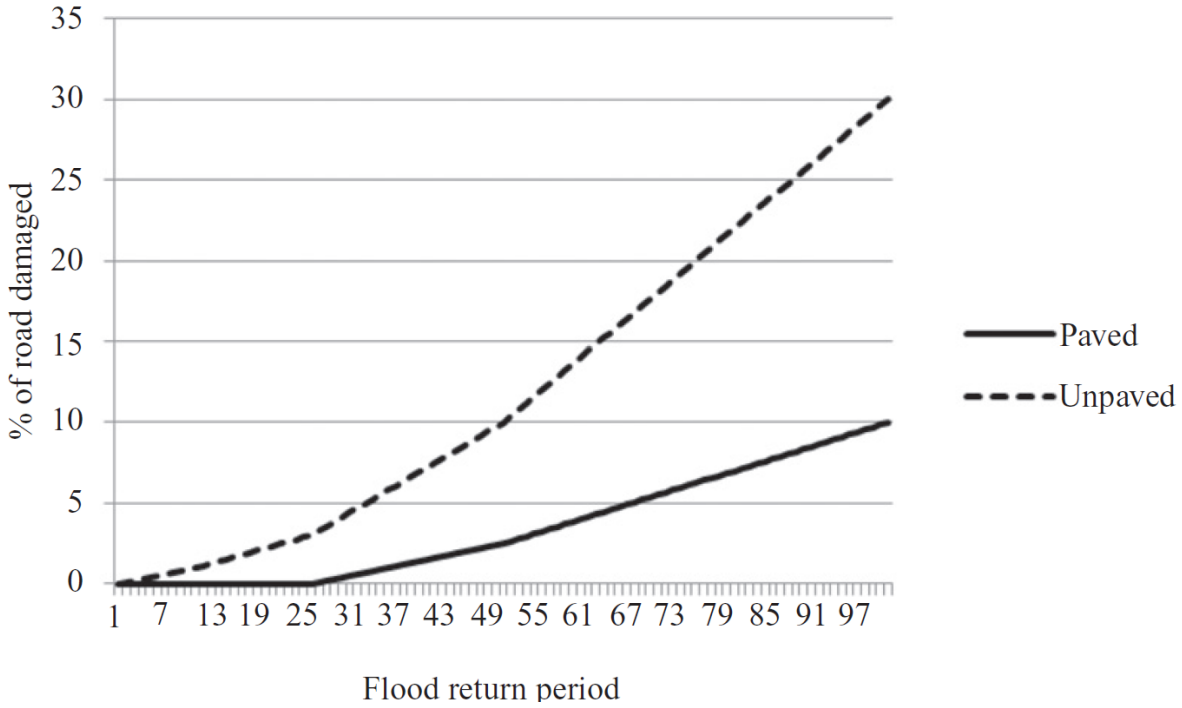


Figure 1.6 Estimated Road Damage Functions by Flood Return Period (Chinowsky, Arndt, 2012).

The graph provides a definite template for determining the financial risk of climatic disasters. It differentiates between minor and major flooding, arguing that smaller ones do not produce any significant damage, whereas a 100-year flood, or an event that occurs with a probability of 1 in a year, is taken as a model with which a large amount of devastation is measured. The resilience of the road surface is used to gauge the vulnerability of the infrastructure. This leads to the assumption that unpaved roads are more vulnerable, and with a perceived rate of damage of 30% in case of a major flood, it is merely 10% in the case of a paved road. This indicates that places with a high density of unpaved roads are much further ahead to recover once a disaster has struck. On the financial aspect, the cost of restoration is not a consistent one. This varies according to the classification of the road (primary, secondary, and tertiary) and material type. Instead of putting a set cost on repairs, the model estimates the cost as a variable percentage of what would have been the cost of constructing new roads. By doing this, the researchers can estimate the "relative" economic burden on an area due to its geography and existing construction standards. (Chinowsky, Arndt, 2012)

1.3 Urban Drainage, Stormwater Management, and Flood Risk

The urban drainage system is one of the most affected infrastructure systems by climate change because it is specifically designed for a set of return-period rainfall statistics that are systematically changing as precipitation intensifies. Loperfido *et al.* (2014) operating in the Chesapeake Bay watershed, clarified that the distributed stormwater best management practices (BMPs) deliver better hydrological performance than centralized systems, as they showcase higher baseflow, reduced small-event runoff, and improved peak attenuation. More importantly, they also concluded that the impervious surface fraction is a stronger indicator of catchment hydrology BMP design, explaining that urban land planning is equally important as drainage engineering to achieve long term hydrological sustainability. The infrastructure-focused drainage adaptation approach is challenged by the positioning of land use planning as a primary resilience solution in addition to standard grey infrastructure modernization.

Zhou *et al.* (2013) offered the most cost-effective analysis of drainage adaptation, which developed an integrated hydrological-economic framework in Aarhus, Denmark. It evaluates pipe enlargement, local infiltration, and open drainage in a green space of the city to the no-action benchmark. The open drainage systems built into recreational spaces are always better than the traditional ones, as it is much better to monetize the entire set of benefits, such as flood control, groundwater recharge, and hedonic amenities with proximity to urban water structures. The hedonic valuation factor, which measures the economic value that the residents assign to an urban area, to the urban water features by examining property prices, transforms the economic argument behind the use of natural based drainage system into an empirical-based cost-benefit analysis. Tajuddeen, Rodrigues (2024) proposed an analysis introducing the social equity dimension and showing that communities with the greatest compound flood risk, high hazard, and low adaptive capacity consistently rank among the most socially deprived groups. The common form of economic risk assessment that prioritizes investment by asset value will therefore lead to systematic under-investment in protecting the most vulnerable group. The research is the sole study among the reviewed articles that argues in physical form and represents a major contribution to the literature that remains largely aggregate-impact-oriented.

1.4 Durability of Structural Materials in Climate Change

The two most complementary studies examine the durability of reinforced concrete under variable atmospheric and thermal conditions. Pedram *et al.* (2022) discussed the effect of increasing the temperature and the concentration of CO_2 in the atmosphere on the two major

concrete deterioration processes, i.e., carbonation-induced depassivation of reinforcement and chloride-induced corrosion. Increased temperature enhances the diffusion rate of CO_2 in concrete cover and speeds up the chemical reaction that occurs beneath it. Elevated CO_2 concentrations enhance the driving concentration gradient, which is effectively additive in nature. Warming also enhances the coastal region, enhancing chloride diffusion in the concrete matrix, while the marine exposure can be experienced through sea-level rise and extensive storm surge. Structures like bridges, ports, and waterfront buildings that are exposed to high CO_2, high temperature, and marine chloride simultaneously can result in compound, synergistic deterioration that can radically reduce service life compared to design assumptions. The paper also examines preventive and remedial solutions, such as increasing cover depth, adding more cementitious materials, and reviewing the surface sealers, and provides viable advice in new designs and in the management of existing assets.

Bastidas-Arteaga, Stewart (2015) extended this mechanistic review with a spatially explicit, probabilistic analysis of climate change impacts on corrosion initiation across Australia. The study is based on a reliability model that uses probabilistic climate forecasts to generate geographically heterogeneous maps of the probability of corrosion initiation at future dates, considering various emission scenarios. Probabilistic framing is also methodologically important to asset management, with the deterioration acceleration expressed as a probability of being structurally deficient at certain future dates. The methodology can be directly applied to risk-based asset management and can be used to make quantitative, rational decisions regarding inspection frequencies and when to invest capital. These two studies will offer a mechanistic conceptualization as well as a quantitative evaluation profile to support climate-informed concrete infrastructure management. The two studies collectively suggest that the current design codes, calibrated based on the historical atmospheric conditions, will become more unreliable in providing the desired service lives and thus that durability requirements in structural design criteria should be systematically revised.

1.5 Tunnel and Underground Infrastructure

Spyridis, Bergmeister (2024) occupy a unique position in the reviewed group because they primarily focus on climate change impacts on tunnels and underground structures, which constitute a growing strategic subset that is scantily covered in the wider literature. The review summarizes geotechnical, structural, and hydrological data, identifying groundwater fluctuation as the dominant climate-related risk. Focuses on both increased hydrostatic pressure caused by increased groundwater recharge and ground movement due to shrinkage, which may lead to

loading conditions larger than those initially assumed, with reported events of both failure modes in several countries. As the temperature on the surface rises, it gradually increases the temperature in the underground structure, which influences the ventilation system activity and increases the rate at which structural lining degrades, with the same mechanisms as those considered by Pedram *et al.* (2022). The strategically important fact is that the tunnel structure has a design life of 100 years or more and will undergo the entire spectrum of the projected climate change during its service life, rendering the previously assumed design with a constant value of geotechnical and hydrological assumptions invalid by default (Spyridis, Bergmeister, 2024). This time dependence argument is partially acute in the case of underground structures, for emergency interventions are much more expensive and cataclysmic as compared to surface structures. The top priority adaptation to the research recommends for existing tunnels is increased monitoring and early warning equipment, which is complemented with climate-based design criteria in new construction. A fundamental weakness is the lack of major relative quantitative studies in this field, currently methodological frameworks to carry out systematic underground infrastructure climate risk evaluation are not as developed yes those to do the same with surface infrastructure.

1.6 Frameworks and Governance of Resilience

Theoretical and applied resilience

Liao (2012) provides the theoretical foundation for the resilience approach that traverses throughout the reviewed literature. The paper contrasts the concept of resilience-based intervention that cities should be designed to withstand, absorb, and recover from flooding with that of resilience-based intervention that aims at avoiding flooding through engineering, and asserts that resistance produces a brittle system that gets disastrously vulnerable to events that go beyond the engineering design criterion. Kabisch *et al.* (2016) provided an empirical extension of the framework with flood management development in the Vietnamese Mekong Delta, which derives transferable design principles that include elevated platforms, flood-tolerant materials, integration of water bodies in urban space, and shows that adaptive coexistence between cities and continuous flooding results in resilience that is cost-effective in practice. These principles directly contrast with nature-based drainage methods that are supported by Loperfido *et al.* (2014) and Zhou *et al.* (2013).

Zhao *et al.* (2022) apply the resilience theory into a four-property engineering concept (robustness, rapidity, resourcefulness, redundancy) to open transformation networks under

climate change, and Zhang *et al.* (2024) generalize it to the multi-sector urban infrastructure, creating a new approach to systems resilience that explicitly considers interdependency, i.e., Failure in one sector cascades to other, and sector-by-sector assessment misses these systematic risks. Collectively, the studies provide the clearest cross-sectoral resilience methodology within the literature reviewed and support the notion that resilience is a property of systems and governance arrangements and not an individual asset.

Governance, Accountability, and Obstacles to Adaptation

The framework that was created by Mees, Driessen (2019) offers the most theoretically elaborated governance analysis in the set reviewed, which has developed a framework to examine the accountability of local climate adaptation arrangements based on 5 mechanisms, i.e., clear mandate, transparency, political oversight, citizen control, and checks and sanctions. The main argument of the paper is that the lack of imprecise or nonexistent accountability structures is a significant and insufficiently studied barrier to adaptation, separate from the more commonly discussed hurdles of finance and technical capacity, restating the adaptation problem as a problem of governance design instead of engineering policies. It is independently concluded by various technical researchers, like Chinowsky *et al.* (2011) identifies design standard revisions as a transport resilience priority, Zhang *et al.* (2022) suggests that lifecycle integrated climate management must be sustained as multi-organizational coordination, Pedram *et al.* (2022), and Spyridis, Bergmeister (2024) suggests a revision in the design codes in the respective sector. An intersection of various domains and approaches contributes heavily to the main findings of Mees, Driessen (2019) The restricting factor in climate change adaptation of infrastructure is governance and accountability, not engineering knowledge.

1.7 Critical Synthesis and Knowledge Gap

Three major convergences come out in the twenty-two studies reviewed.

1. Everyone agrees that climate change is inflicting actual, increasing, and comprehended stress on civil infrastructure, and that stress will escalate across every possible emission pathway; there is no disagreement on the direction of this observation.
2. Any research dealing with adaptation always finds that there are technically effective responses in all sectors, but they are not being pursued on the scale and speed necessary, and the gap lies on an institutional level, as well as financial and political, more than scientific or engineering.

3. What transition of resistance-based to resilience-based design strategies is supported by the researchers across all sectors, acknowledging that no design standards can ensure protection against every future scenario in the changing climate, and physical protection would be enhanced by accommodating the ability to absorb, recover, and adapt.

Review literature even deals with the productive tensions. The most important is the technically oriented literature, like the work of Bastidas-Arteaga, Stewart (2015), Dikanski *et al.* (2017), Pedram *et al.* (2022), Pregnolato, Ford, Glenis, *et al.* (2017b), and Pregnolato, Ford, Wilkinson, *et al.* (2017a), which gives detailed engineering advice but mentions relatively little about the circumstances of implementation and the studies that are governance-oriented. The adoption of these perspectives has been an issue that is yet to be solved. The second conflict consists of the economic rationale of adaptation, examined by Neumann *et al.* (2015) and Zhou *et al.* (2013) as convincing in the long run, and the institutional barriers in politics and the economy to making these investments, as Mees, Driessen (2019) and Liao (2012) make clear.

Many knowledge gaps need to be prioritized for future studies. The interdependence between infrastructure systems under climate stress, an issue tackled by Zhao *et al.* (2022) and Zhang *et al.* (2024), should be quantified more comprehensively. Only two studies (Chinowsky, Arndt, 2012; Tajuddeen, Rodrigues, 2024) directly tackled the equity dimension of climate infrastructure effects, which resulted in a disproportionate amount of the literature on aggregate economic consequences. The topic of underground infrastructure is carried out with the consideration that very little primary research has been done about this set area of work (Spyridis, Bergmeister, 2024). The set has little evidence on comparative case studies of what works in terms of governance arrangements in providing infrastructural adaptation at scale, and not what good governance theoretically should be. Lastly, the active behavior of infrastructure when considered under compounding of multiple climate stressors over entire multi-decadal lifetimes, and not at time-slices of narrow future expressions, should receive specific modeling research.

1.8 Comparative Analysis: Baltic States vs. EU Benchmarks

The economic cost of climate-related damage is hard to quantify because the record of damage seldom separates different causes (including those climate-related) and rarely takes into account indirect costs (traffic disruption, business losses, emergency response). The most notable point that can be noted was that by the year 2050, the total cost of infrastructure damage would have increased by a further 50-80 % than the costs exhibited in the first two decades of the 21st century (European Environment Agency, 2017).

To put the perspective of infrastructure risk associated with climate change in the Baltic states, it is informative to compare the main indicators of vulnerability with EU standards. A summarized comparison using European Environment Agency (EEA) and Latvian Environmental, Geology, and Meteorology Centre (LVGMC) data can be found in Table 1.1.

Table 1.1
Comparative climate-infrastructure vulnerability indicator (European Environment Agency, 2024; Līdzšinējo Un Nākotnes Klimata Pārmaiņu Rīks : Klimata Portāls, n.d.)

Indicator	**Latvia**	**Baltic Average**	**EU-27 Average**
Warming rate (1961-2020) (˚C)	+ 1.4	+ 1.3	+ 1.0
Increase in extreme precipitation events (%)	+ 20	+ 18	+ 14
Bridges at flood risk (%)	23	19	12
Urban drainage at-risk area (%)	65	62	48
Coastal erosion risk (km coastline)	High (498 km)	Moderate-High	Variable
Infrastructure climate-damage cost (€/km/yr)	~ 4,200	~ 3,800	~ 3,100

The data in Table 1.1 confirms that Latvia and the Baltic states are vulnerable to all major indicators, due to the combination of above-average warming rates, ageing of infrastructure stock, and increased proportional exposure of the transport and drainage assets to climate-related hazards. The results justify the focus of specific policies and engineering investments in climate resilience in the Baltic infrastructure.

2. Theoretical Part

2.1 Conceptual Foundations of Climate Risk in Infrastructure

The IPCC risk framework

The Intergovernmental Panel On Climate Change (IPCC) (2023), Sixth Assessment Report (AR6) states that climate risk is a combination of three interacting elements, i.e., hazard, vulnerability, and exposure. As per civil infrastructure:

1. Hazards are physical climate-related events, such as floods, heat waves, storms, sea level rise, etc.
2. Vulnerability is the susceptibility of the infrastructure to be negatively impacted, both in terms of Sensitivity to the hazard as well as its adaptive ability.
3. Exposure can be considered the availability of infrastructure assets in places where the hazards take place.

This three-part system forms the basis of the thesis, as it enables the systematic degradation of infrastructure risk and identifies recommendations for specific adaptation interventions. The overall risk can be reduced by tackling any of the three components, such as reducing exposure by rerouting infrastructure out of flood-prone areas, reducing risk by enhancing drainage capacity, and reducing vulnerability through strengthening infrastructure against flood loadings.

Representative scenarios found on the Shared Socio-Economic Pathway (SSP) are applied throughout the IPCC report: SSP1-1.9, SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5. These scenarios represent the most extensive inventory of greenhouse gases and future air pollutants, and include high and low CO_2 emissions pathways, with no mitigation of private change occurring. In these scenarios, differences in air pollution control and differences in mitigation strictness in climate change are important influences on the anthropogenic emissions pathways of SLCFs. Studies of modelling, using Representative Concentration Pathway (RCP), complement the assessment based on SSP scenarios, such as at the regional scale.

The purpose of Figure 2.1 is to demonstrate the process chain starting from anthropogenic emissions to alterations in atmospheric concentration and earth energy balance (forcing), to change global climate, and finally, regional climate and climatic impact drivers. The demonstrated figure contains a core set of five Shared Socio-Economic Pathway (SSP)

scenarios, and the emissions and concentration ranges along with the prior Representative Concentration Pathway (RCP) scenario in the year 2100.

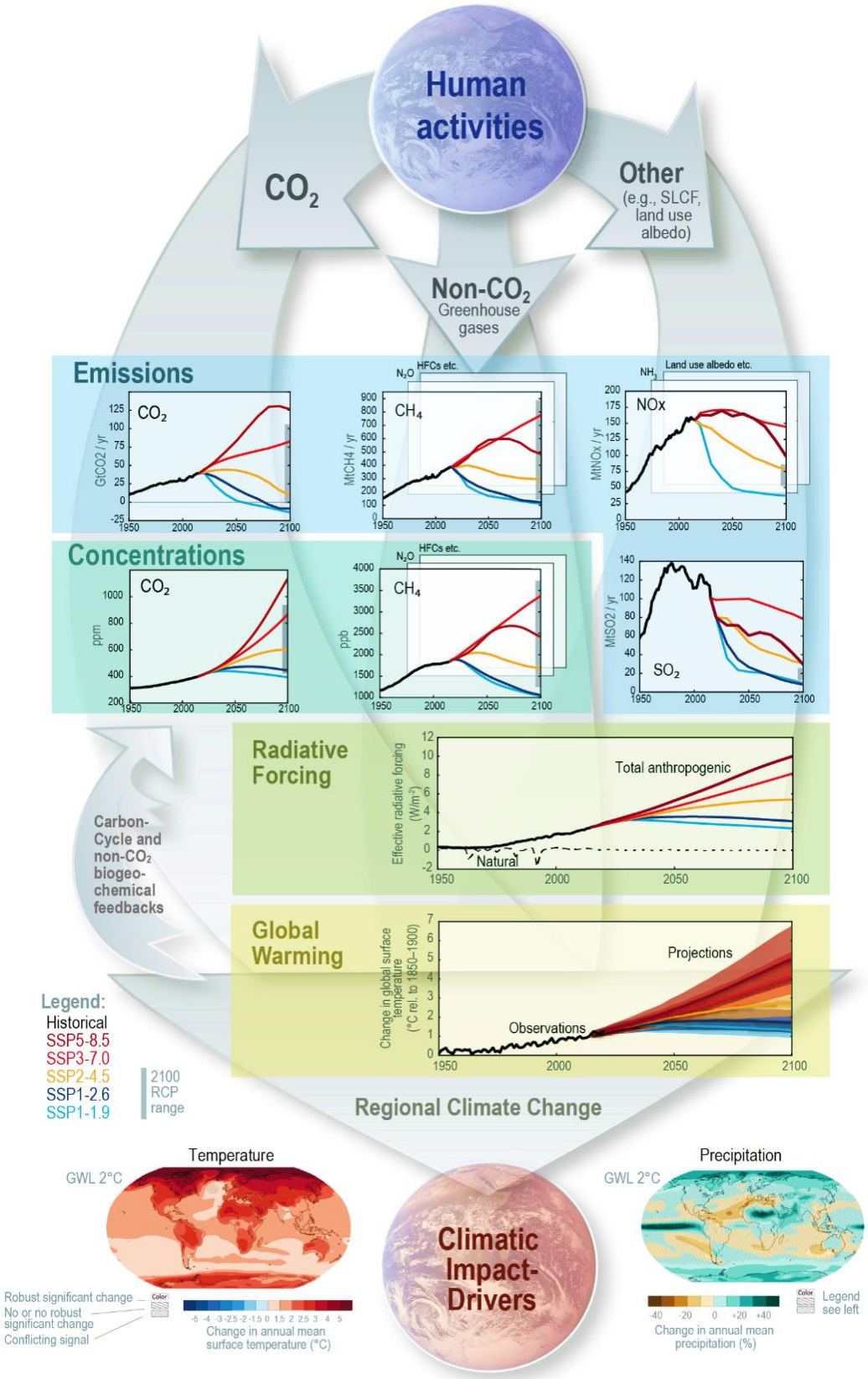


Figure 2.1 The climate change causes and effect chain (Intergovernmental Panel on Climate Change (IPCC), 2023).

The demonstrated figure contains a core set of five Shared Socio-Economic Pathway (SSP) scenarios, and the emissions and concentration ranges along with the prior Representative Concentration Pathway (RCP) scenario in the year 2100.

Critical Infrastructure and Resilience Theory

The topic of resilience is becoming more prominent in infrastructure engineering and in policy decisions. In the context of climate adaptation, infrastructure resilience can be defined as the capacity of a system to absorb climate-related stresses, continue to perform core functions despite disruptions, and quickly return to its pre-event performance level. System resilience is not simply about the ability to resist disturbance, but it is also the ability to absorb disturbance, to adjust in altered conditions, and change when required to sustain important activities. Resilience under socio-economical term involves learning, reorganizing, and preserving social-economic identity through change.

Liao (2012) developed an urban resilience to flood theory that explicitly compares a resilience-based approach, where the urban system is built to tolerate, absorb, and recover from flooding, and engineering intervention targets to eliminate flooding. The theory argues that when resistance strategies work well, responding to the events within the design capacity, it produces dangerous rigidity. Once a resistance system is broken by an extreme event that is beyond the design standards, the results are often Disastrous then they would have been under a more controlled system. This observation has been highly impactful in flood risk management in cities and has direct counterparts in other infrastructure sectors, where extreme-event designs are being used more commonly than means-condition designs.

Kabisch *et al*. (2016) extended the framework analytically to study communities experiencing seasonal flooding in the Vietnamese Mekong Delta, obtaining urban design principles that ensure flood resilience based on ecological knowledge reflected in the local flood management behaviors. They showed that communities that plan urban space to be flexible to the periodic flood by accepting it occasionally and not to oppose at, are the ones that have greater adaptive capacity and social resiliency over the years, with implications for planning new urban development designs in flood-prone areas across the world.

From a civil infrastructure perspective, technical resilience is the most applicable and includes the provision of redundancy (having backup plans when primary systems are lost), robustness (structural strength against extreme loads), rapidness (how quickly a structure recovers), and resourcefulness (capability to mobilize resources upon disruption). These principles guide the practical suggestions made in Chapter 3.

Adaptive management and infrastructure lifestyle

Conventional infrastructure planning uses a predict-then-act model in which engineers make predictions about future demand and environmental conditions and then design and construct to fulfill operational requirements. Climate change challenges this model by introducing profound uncertainties into long-term predictions. In comparison, adaptive management is an iterative methodology where decisions are made progressively, monitoring is used to track ongoing changes, and flexible designs are made to address the scenarios that may occur in the future.

When applied to civil engineering infrastructure, adaptive management suggests that structures must be designed with the possibility to make upgrades in modules, establishing decision triggers such as a set point in the rise in sea level above which a pre-determined flood barrier is installed, and the establishment of monitoring mechanisms that give early notifications of a nearing threshold. This strategy is alternatively known as adaptation pathways planning, as described by Haasnoot *et al.* (2013) and followed by numerous national adaptation plans in Europe.

2.2 Regulatory and Standards Framework

Eurocodes and Climate Consideration

Eurocodes are the standard European structural design rules that regulate the design of buildings, bridges, geotechnical structures, and other civil engineering constructions across the EU. The primary Eurocodes applicable in climate adaptation are EN 1990 (basis of structural design), EN 1991 (actions on structures), EN 1997 (geotechnical design), EN 1998 (Seismic Design) Aye numerous material-specific codes (EN 1992-1999).

Historically, Eurocodes have always included climate actions, such as wind, snow, and terminal actions, but based on structural analysis of historical climate data, normally scaled to 50-year return periods. But the Eurocodes of the second generation (In continued development according to the directive of the European standardization committee, CEN) are adding clear references such as climate-adjusted return period and extension of design service-life under future projections.

EN 1991-1-5 (Terminal actions) is specifically relevant to the infrastructure design in the Baltic context. This code identifies pommel action inputs, including maximum and minimum shade air temperature, bridge temperature distribution, based on national meteorological data. The design characteristics of the maximum shade air temperature should be revised to depict

anticipated conditions at the end of the design service life, since climate projections show constant warming at all locations. The anticipated value could be higher than the observed values in the Baltic Region in 2070-2080.

ISO Standards for Climate Adaptation

The International Organization for Standards (ISO) has developed a set of standards addressing adaptation to climate change. The ISO 14090 (2019, Adaptation to Climate Change – Principles, Requirements and Guidelines) offers a systematic approach through which organizations, such as infrastructure owners and operators, may evaluate significant climate risks, define adaptation opportunities, and engage in adaptation actions. The framework is further extended to ISO 14091 (2021) for vulnerability, impact, and risk management.

The ISO 14090 process is a five-step process that the infrastructure manager should follow sequentially

1. Understanding the organization and the surrounding climate settings
2. Evaluating risks and opportunities related to climate
3. Formulating strategies and plans for adaptation
4. Implementing adaptation measures
5. Monitoring, evaluating, and reporting

This iterative, cyclical process is directly related to the adaptive management model described earlier.

EU Adaptation Strategy and Policy Context

According to the strategy on adaptation to climate change (2021) by the European Commission, the EU desires to transform Europe into climate resilient by 2050. The strategy has identified 4 main goals:

1. To make adaptation smarter (enhancing the evidence base)
2. to make adaptation faster (increasing the rate of adaptation measures implementation)
3. making adaptation more systematic (adapting a central part of all the relevant policies)
4. To step up international action on adaptation

Under civil infrastructure, operationalization of the strategy is carried out through the EU Taxonomy Regulation, which requires that new infrastructure projects demonstrate that they meet minimum climate adaptation requirements. European Union-funded projects under the cohesion fund or the Connecting Europe Facility (CEF) are required to be climate-proofed, that is, involving A systematic process of evaluating the impact of climate risk throughout the

project life span and the implementation of relevant adaptation strategies. Latvia, as a significant recipient of EU structural funding in infrastructure, comes under such requirements.

In 2019, the Latvian national adaptation plan identified transport infrastructure, coastal protection, and water management as the priority areas for climate change. The proposal includes changes in national technical norms on infrastructure design that regulate infrastructure design to include climate projections, in the area where minimal progress has been made so far.

2.3 Climate Risk Assessment Methodology for Infrastructure

Quantitative Risk Assessment

Quantitative climate risk assessment (QCRA) of infrastructure is a process that uses numerical estimates of the likelihood and impact of the occurrence of climate-related failure or degradation. The basic correlation can be observed in Formula (2.1) (Intergovernmental Panel On Climate Change (IPCC), 2023).

$$R = H \times E \times V \times C, \tag{2.1}$$

Where R – Risk;

H – Hazards;

E – Exposure;

V – Vulnerability;

C – Consequences.

Where each term is measured using appropriate data and models. Climate model outputs (example changes in 100-year flood discharge) are generally used as hazard characterization but are often scaled down (statistically or dynamically) to a regional or local scale. Exposure measures are evaluated via asset inventory and Geographic Information System (GIS) analysis. Vulnerability is defined in terms of functions of fragility, empirical or modeled relationships between the intensity of hazards and the state of infrastructure damage. The repair cost database or traffic disruption models estimate the cost of consequence, measured in economic terms.

Flood Return Period Analysis and Nondimensional Flow Coefficient

In hydraulic infrastructure (bridge, culvert, drain system, etc.) A key parameter in a climate risk assessment is the design discharge based on a certain written period. The rational

method gives a general-purpose first-order estimation of peak runoff in a storm, which is given by Formula (2.2) (Intergovernmental Panel on Climate Change (IPCC), 2023):

$$Q = C \times i \times A, \tag{2.2}$$

Where Q – peak discharge (m^3/s);

C – Dimensionless runoff coefficient (0.3 to 0.9, depending on the land use and the soil permeability);

i – Rainfall intensity over the design period (m/s);

A – catchment area (m^2).

The most important variable in climate change is the rainfall intensity I, to which there will be an upsurge in the specific return. As the climate forms in the case of Baltic states, projections suggest an increase in 100-year return period rainfall intensity of 10 to 20 % by 2070 under the scenario of RCP 4.5.

This has a direct impact on the design of a drainage system, including a culvert or drainage channel, with the size that carries the current 100-year flood. Might only be able to carry the existing 50-year flood under the immediate climate change projection, essentially due to an increased likelihood of an overflow condition.

Thermal Expansion Analysis in Bridge Design

Bridge deck thermal movement is an essential design factor. The bridge deck thermal movement coefficient of thermal expansion (α_T) is regulated by the temperature range in which the structure operates. The change of ΔL of a bridge deck of length L over a change of temperature ΔT of the bridge deck is given in Formula (2.3) (Intergovernmental Panel On Climate Change (IPCC), 2023):

$$\Delta L = \alpha_T \times \Delta T, \tag{2.3}$$

Where ΔT – Change of temperature (˚C);

α_T – Coefficient of thermal expansion – Structural Steel = 12×10^{-6} /˚C;

– Concrete = 10×10^{-6} /˚C.

Maximum air temperatures increased because of climate change, resulting in increased demands on expansion joints and bearing systems.

In Latvia, the effective bridge temperature for steel bridges is historically considered between -30˚C and +40˚C. Under Representative Concentration Pathway (RCP) 4.5 projections

for the 2070 design horizon, the upper limit can rise to +43˚C to +46˚C, increasing the expansion joint capacity by 7 to 12 %.

This small modification has a practical advantage, as the expansion joints and bearings are the elements demanding the highest maintenance. Extending the design temperature range may require joint replacement or structural modification of an existing bridge, with fixed or near-capacity expansion.

Scour Risk Assessment

The most common cause of bridge failure all over the world, which is directly connected to the amount and duration of the flood, is bridge scouring. It is the erosive removal of bed material surrounding the bridge foundation by the flowing water. The maximum scour depth at the bridge pier (y_s) can be calculated by using the Colorado State University (CSU) equation (Saha *et al.*, 2018),

$$\frac{y_s}{y_1} = 2.0 \times K_1 \times K_2 \times K_3 \times K_4 \times \left(\frac{a}{y_1}\right)^{0.65} \times {Fr_1}^{0.43}, \quad (2.4)$$

Where
- y_1 – Flow depth (m);
- a – pier width (m);
- Fr_1 – Froud number of the approach flow;
- K_1 – correction factor for pier shape;
- K_2 – correction factor for flow angle;
- K_3 – correction factor for bed condition;
- K_4 – correction factor for armoring.

This equation implies that even a 20 % increase in the peak discharge can relatively increase up to 10 to 15 % of the design scour depth, further requiring stronger foundation protection.

Frost Index and Penetration Depth

The Frost Index (FI), or the sum of a winter of degree-days below 0˚C, is traditionally used as a measure of how badly infrastructure would have fared during a frost season. The more Frost Index, the longer and more intense the frost period. The Stefan equation is defined in such a way that the relationship between the Frost Index and the depth of frost penetration in the ground requires the described function of the Stefan equation (Formula 2.5) (Freitag, McFadden, 1997):

$$Z = \sqrt{\frac{2 \cdot \lambda \cdot FI}{\rho \cdot L}}, \tag{2.5}$$

Where Z – frost depth (penetration depth) (m);

λ – thermal conductivity of the frozen soil (In some modified versions, λ is a dimensionless correction factor, and k is used for conductivity) (W/(m.K));

FI – Frost Index, which represents the cumulative magnitude and duration of below-freezing temperatures;

ρ – Dry density of the soil (kg/m^3);

L – Latent heat of fusion per unit volume, which is the energy released when the water in the soil pores freezes.

For the Latvian pavement subgrade condition (Sandy Loam soil – λ = 1.8 W/mK, ρ = 1750 kg/m^3) (*Līdzšinējo Un Nākotnes Klimata Pārmaiņu Rīks : Klimata Portāls*, n.d.). Therefore, the equation simplifies into $Z = 0.032\sqrt{FI}$.

Table 2.1

Historical and projected Frost Index and penetration depth values for Latvia (*Līdzšinējo Un Nākotnes Klimata Pārmaiņu Rīks : Klimata Portāls*, n.d.)

Period	**Frost Index FI (degree-days)**	**Frost Depth Z (m)**	**Estimated Design Frost Layer (m)**	**Change vs. 1961-1990**
Historical baseline 1961-1990	810	0.91	0.95	-
Recent observations 2001-2022	685	0.84	0.88	- 7.7 %
2050, RCP 4.5 median	550	0.75	0.79	- 16.8 %
2070, RCP 4.5 median	430	0.66	0.70	- 26.3 %
2070, RCP 8.5 median	310	0.56	0.59	- 37.9 %
2100, RCP 8.5 median	180	0.43	0.45	- 52.6 %

Table 2.1 shows how the Frost Index and the resulting frost penetration depth have changed and are projected to change in Latvia. It concludes that warmer winters are positive news to infrastructure - the less the frost penetration, the less thick the needed pavement structures, and the less the frost heave.

That would be an incorrect conclusion, for the following reason. It is not only the depth of frost that causes pavement damage, but rather the frequency of movement of the temperature across the 0˚C threshold in a season of winter known as freeze-thaw transitions. Every transition cycle will cause the water in the pavement pores to dilute (when freezing) and constrict (when thawing), which successively breaks up the material structure. An intensive, permanent frost action is indeed less destructive than a light, but rapidly alternating action, because in the former the pavement becomes solidly frozen, whereas in the latter it becomes liquidly frozen and liquidly unfrozen at the same time.

Latvian Environmental, Geology, and Meteorology Centre (LVGMC) records show that the number of freeze-thaw transitions per winter season in Riga has increased from an average of 28 in the period 1961-1990 to an average of 41 in the period 2001-2022, a 46 % increase. It is likely to carry on as winters grow warmer: there will be more days alternating between the freezing point and periods when the deep frost is maintained. Extending this correlation to the 46 % rise in the frequency of transition observed in Latvia suggests that the effective pavement life in the most exposed sites in Latvia would be reduced by about 34 %. A 34 % increase in the rate of deterioration is another liability based on the maintenance costs.

2.4 Adaptation Strategies and their Economic Evaluation

Classification of Adaptation Measures

The climate-vulnerable infrastructure adaptation practices are characterized in two main types: temporal (Anticipatory vs. Reactive) and structural (Technical vs. Non-structural). Anticipatory measures are taken before the event, in advance of projected climate impacts being observed, and are usually cheaper than reactive measures implemented after the damage. Technical measures are the physical changes in the infrastructure, whereas non-structural measures include regulatory, institutional, and informational reforms.

The adaptation options in Latvia and the Baltic States for the civil infrastructures:

1. Technical – Anticipatory: Improving drainage capacity, reinforcing the foundation of bridges, reconstructing high-risk road areas, construct coastal resisting structures.
2. Technical – Reactive: Emergency repairs like slope stabilization, pavement surfacing after floods and storms.

3. Non-structural – Anticipatory: Updating design standards for tackling climate change, identifying flood-prone zones to restrict development, and developing monitoring systems across the area.
4. Non-structural – Reactive: Emergency quick response plans, insurance support.

Cost-Benefit analysis framework for adaptation

Economic analysis done for the adaptation practice is usually done through cost-benefit analysis (CBA), where the cost allotted for adaptation measures is compared with the benefits of avoiding damages during the planning period, where the cost is properly discounted to present value. The net present value (NPV) of an adaptation measure is shown in Formula (2.6), a fundamental metric used to determine the profitability of the project (Hopkinson, 2024).

$$NPV = \sum_{t=1}^{T} \frac{B_t - C_t}{(1+r)^t} - I_0, \tag{2.6}$$

Where B_t – Benefits (Cash inflows) received at time t;
C_t – Operating costs (Cash outflows) incurred at time t;
r – Discount rate representing the opportunity cost of capital;
t – specific time period (years);
T – Project's lifespan (years);
I_0 – Initial investment at t = 0.

If NPV > 0, it is a good investment as the project is expected to generate value above the cost of capital. Whereas if NPV < 0, it is typically not accepted as the project outcomes a net loss relative to the discount rate. To ease the infrastructure adaptation in the European Union, a social discount rate of 3 to 5 % is applied, recommended by the European Commission's Guide to Cost-Benefit Analysis of Investment Projects.

The Global Commission on Adaptation (2019) published a report mentioning that an investment of USD 1.8 trillion worldwide between 2020 and 2030 in adaptation will produce a benefit of USD 7.1 trillion, showing the Cost-Benefit ratio of 1:4.

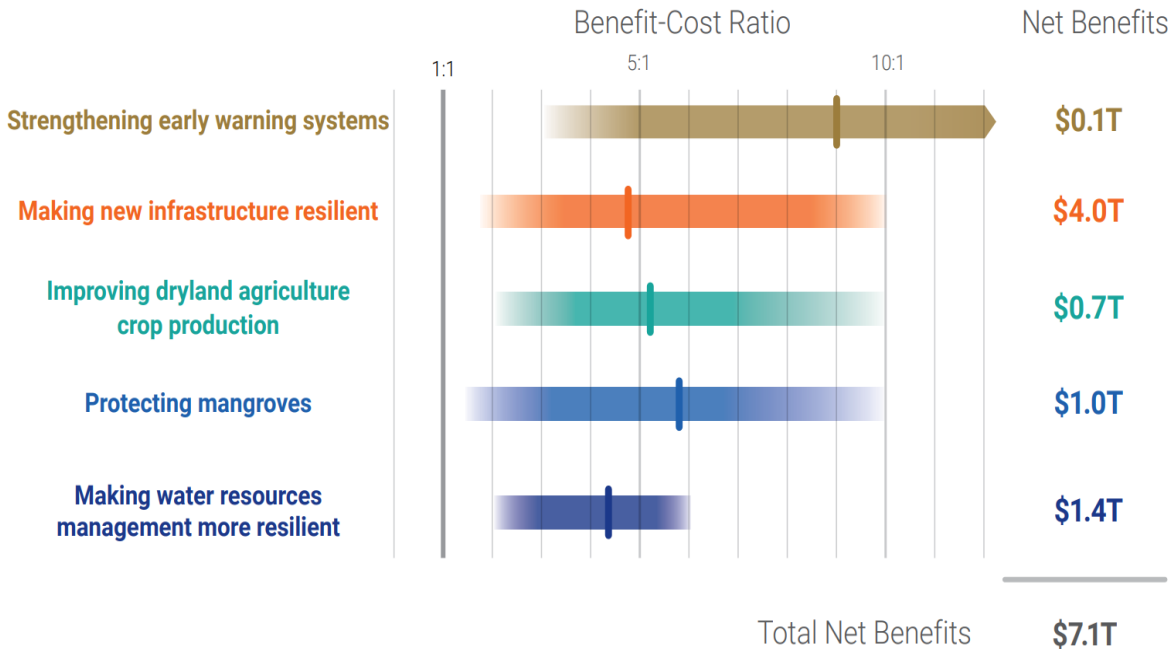


Figure 2.2 Benefits and Costs of Illustrative Investments in Adaptation.
(The Global Commission on Adaptation, 2019)

The idea of the figure is to demonstrate the macroeconomic argument that there should be investment in a variety of adaptation strategies. The net benefits illustrate the approximate global net benefits to be gained by 2030 from an illustrative investment of $1.8 trillion in five areas (the total does not equal the sum of the rows due to rounding). The appearance of the actual returns is determined by numerous factors (including the growth of the economy and its demand, the context of the policy, institutional functioning, and the state of assets). Also, these investments do not address everything that may be needed across sectors (e.g., adaptation in the agricultural sector will involve much more than just dryland crop production) or even all sectors (as the health, education, and industry sectors will not be part of it). This graphical representation should not suggest the complete comparability of the investments in various sectors or countries due to the data and methodological limitations.

Lifecycle Cost Analysis

When deciding on the adaptation, it is necessary to evaluate both the initial investment cost and the infrastructure's entire lifespan, including maintenance, repair, and rehabilitation costs, as well as the residual value at the end of its life. Lifecycle cost analysis is expressed in Formula (2.7) (Zia, 2020).

$$LCC = I_0 + \sum_{t=1}^{T} \frac{M_t + R_t + D_t}{(1+r)^t} - \frac{S_T}{(1+r)^T}, \tag{2.7}$$

Where I_0 – Initial investment at t = 0;

M_t – Maintenance and operating Costs;

R_t – Replacement Cost;

D_t – Disposal/Demolition Cost at the end-of-life span;

S_T – Salvage Value;

r – Discount rate;

$(1 + r)^t$ – Discount factor.

With climate change, D_t rises gradually during the time of planning phase. It happens as the frequency and the severity of the climate hazards increase. When considered ahead of 2050 or later, it becomes economically smarter to invest early for adaptation rather than spending more on repair work.

2.5 Green and Natural-Based Infrastructure Solutions

Nature-Based Solution (NBS) refers to a growing category of climate adaptation strategies in which ecological processes are used to reduce the vulnerability of infrastructure against the hazards associated with climate change.

Natural-Based Solution in the context of the Baltic states is as follows:

1. Use of Urban green infrastructure, such as permeable pavements, green roofs, rain gardens, and urban wetlands, to minimize stormwater runoff and urban heat island effects.
2. Restoring Coastal and Riparian Vegetations to reduce erosion, dissipate wave energy, and stabilize embankments.
3. Through the construction of stormwater retention infrastructure and treating wetlands, the flood storage capacity was increased.
4. Native vegetation deep-rooted through bioengineering to stabilize slopes can become an alternative to conventional geotechnical retaining structures.

The EU Biodiversity Strategy for 2030 (European Commission. Directorate General for Environment., 2021) suggests supporting NBS by incorporating a law that promotes its inclusion in the planning and governance of infrastructure. Along with climate adaptation, NBS provide co-benefits for biodiversity enhancement, improvement of water quality, and increasing recreational and amenity values. With all the co-benefits, if properly monetized, NBS can be considered a good cost-benefit solution compared to engineering solutions only.

2.6 Digital Tools and Monitoring Technologies

The combination of digital monitoring, sensing, and modelling technology helps to improve infrastructure management to detect, predict, and respond to the risks associated with climate. Key technologies include:

1. A structural health monitoring (SHM) system consists of embedded sensors to continuously measure the strain, vibration, temperature, and displacement of the structures to provide early warning of climate-induced damage.
2. IoT-enabled flood forecasting Network monitoring real-time rainfall, river level, and soil moisture data to prevent flood-like situations at the catchment scale.
3. Geographical information system (GIS) and remote sensors analyzing infrastructure exposure and vulnerability, including LiDAR-based terrain modeling to map flood streams.
4. Introducing building information modeling (BIM) and digital twin systems with the ability to test climate adaptation and optimization of maintenance solutions.

The European Drought Observatory and Copernicus Emergency Management service, and other digital infrastructure monitoring systems developed by the joint Research Center of the European Commission, enable the Baltic state authorities to access climate risk information at minimal marginal cost. These platforms can be used to integrate with national infrastructure management systems as a high-value, low-cost adaptation measure.

2.7 Systematic Interdependencies and Cascading Failures

One important aspect of climate risk in civil infrastructure is that it is commonly underestimated that all the analyses conducted within the sector are mutually reliant. The contemporary urban and regional infrastructure, like transportation, water, energy, and telecommunication are so highly interdependent that failure of one system can trigger a domino effect in failing the other systems. A good example to explain it was the flooding of the Latvian highway P2 near Ogre that happened in 2010, not only disrupting the road traffic, but it also affected the access to the emergency services, affected the supply chains to local businesses, and isolated the communities without necessary services. The important interdependencies between different sectors, which enable climate change impacts, are shown in Figure 2.3.

Two new methodological tools to measure systematic interdependency risk in infrastructure systems include agent-based modelling and Network topology analysis. These

methods describe infrastructure elements as nodes in a graph, and dependencies as either physical or functional parts. By removing the nodes to simulate component failure, it is possible to propagate the spread of failure throughout the system and identify critical nodes whose failure has a systematic disproportionate effect.

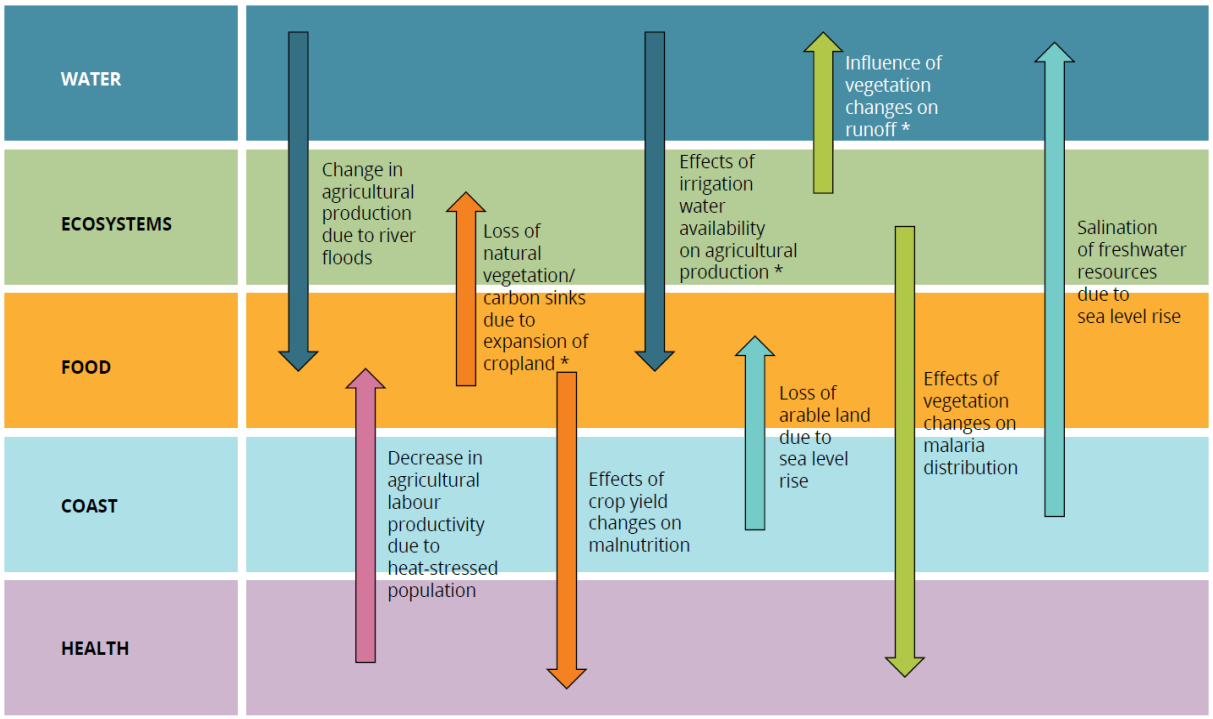


Figure 2.3 Examples of interdependence between climate impacts on different sectors (European Environment Agency, 2017).

This figure represents the underlying logic of a cross-sectoral assessment by schematically mapping the complex interdependencies and feedback loops among climate-sensitive systems. The cross-sectoral approach compares existing data on multiple sectors simultaneously to determine this or that hotspot of vulnerability. As the figure illustrates, when any single climate driver, such as Water (such as river flooding), affects single climate drivers like Food (agricultural output) and Health (malnutrition), single climate drivers can directly affect the other climate drivers. This visual web points out that although modeling in such interdependency adds to the technical complexity of the system, it is necessary to model in such interdependency to attain a realistic perception of how climate change impacts and adaptation policies operate within a larger socio-economic framework.

Types of Interdependencies

Based on the framework that Rinaldi, Peerenboom, and Kelly (2001) developed, the infrastructure interdependencies can be clustered into four groups:

1. **Physical interdependencies:** a system relies on the physical production of the other system. An example of this is that water treatment plants require electricity; access to roads is needed to supply some form of fuel to heating systems.
2. **Cyber interdependencies:** one system is dependent on information or control signals from another system. SCADA in water networks relies on power grids; traffic systems rely on telecommunications.
3. **Geographic interdependencies:** two or more systems fall in the same geographic space, thereby meaning that a single hazard impacts everyone at the same place in time. An example of some of the objects impacted by a riverside flood includes the road, the railway, the utility trench, and the influx of water on that corridor.
4. **Logical interdependencies:** the failure of one system influences another through an economic or social channel. Businesses close due to damage of road, and this loss of business requires the government to maintain the roads, thereby causing more damage.

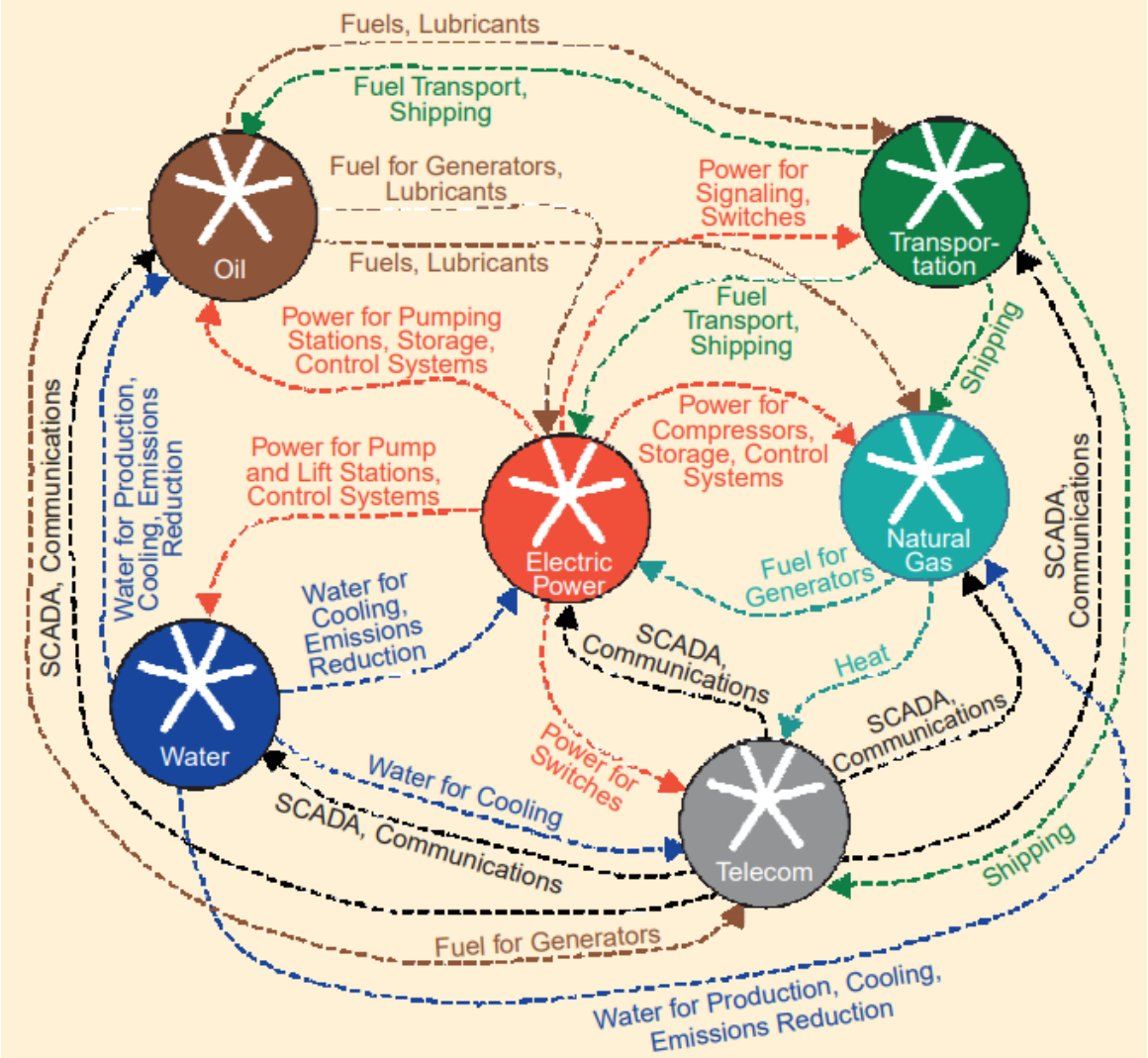


Figure 2.4 Infrastructure Interdependencies (Rinaldi *et al.*, 2001)

The figure of the infrastructure interdependency model shows interdependency and that the six critical sectors of the model, Electric Power, Oil, Natural Gas, Water, Telecom, and Transportation, are not independent entities but an interdependent system, as it depends on each other and vice versa. At the heart of this web lies the Electric Power that happens to be a key enabler by supplying the energy needed to drive pumps, control systems, and signaling in nearly all other application domains. This forms necessary feedback loops, including the Power-Water Nexus, whereby electricity drives water to the wider world, and water is simultaneously needed to cool and reduce emissions in power plants. Likewise, the alteration of the digital physical nexus of the relationship between Power and Telecom suggests the idea of dependence on a stable amount of power to stay in operation.

In addition to these technical loops, the model illustrates the logistical support that necessarily signifies the physical transportation of lubricants, shipping services, and generator fuels to keep facilities operational. The diagram highlights, by mapping these cross-sectoral connections, that a failure in one node, such as a localized power outage or a telecommunications failure, can have a cascading effect, impacting the resilience of the entire regional infrastructure. The view is core to engineering and policy planning, as the perspective shifts the vantage-point away from an individual asset maintenance vantage-point, in favor of a systemic protection of so-called weak points within the system vantage point, otherwise allowing the system to be subject to widespread service failures.

Also, the diagram emphasizes that contemporary infrastructure is becoming increasingly smart, i.e., the layer of digital communications (Telecom) is becoming as important as the very process of water or fuel. During a crisis, because data transmission between SCADA systems is not possible, the response teams will be blinded to the effect, even if a physical repair resource can soak up the excess water. This means that defense in these areas should consider a comprehensive security approach that should not only consider physical threats, such as extreme weather, but also other cyber vulnerabilities that may cut the communication cords holding the entire web together.

2.8 Governance, Policy, and Barriers to Adaptation

Climate resilient infrastructure strategies of technical adaptation can only be achieved with the structure of proper governance to Commission, mandate, finance, and implement them. Governance of climate adaptation of infrastructure is multifaceted, cutting across multiple governmental levels, multiple areas of institutional activities, multiple professional fields, and various time scales. The main obstacles to effective adaptation under a governance context are

institutional fragmentation, capacity issues, and political economy challenges (Mees, Driessen, 2019).

Table 2.2

Overview of the management challenges related to vulnerabilities to climate change for four European regions (European Environment Agency, 2017)

Topic	Northern Europe and the Arctic	North-western Europe	Central and Eastern Europe	Southern Europe and the Mediterranean
Management of urban areas				
Management of rural settlements				
Management of energy consumption in housing				
Management of hydropower production				
Power production with boilers				
Water management				
Management of agriculture				
Forest management				
Coastal management				
Management of health care				
Biodiversity management				
Development and diffusion of green innovations				

Legend	
	Mainly positive opportunities
	Balance between negative challenges and positive opportunities
	More negative challenges than positive opportunities
	Mainly negative challenges

The first major issue in governance is the mismatch between the time scale of political decision-making, generally a four- to five-year election cycle, and the time scale of climate adaptation in infrastructure, which is sometimes decades or centuries long. The benefits of investments in climate adaptation are mostly realized in the future, usually after the political term of the decision-makers who sanctioned them. This creates insufficient political incentives, which demotivates adaptation and postpones difficult decisions. This misalignment in the time spans is found to be one of the most significant structural impediments in investment allotments for climate adaptation in both developed and developing countries. (Chinowsky, Arndt, 2012),

particularly in developing countries where short-term development is prioritized over long-term adaptation requirements.

International environment and policy frameworks

At a global stage, the Paris agreement targets to restrict global warming 2 levels much less than 2˚C, which offers a general framework to both mitigation and adaptation. The adaptation provisions of the agreement, however, are not as prescriptive as its mitigation targets: the parties must negotiate their adaptation priorities and participate in creating adaptation plans, though there is no mechanism by which it becomes obligatory to ensure that the adaptation investments are adequate to avoid the estimated damage. The assessment conducted by the European Environment Agency (EEA) indicates that, although European Union member states have made significant progress in developing national adaptation strategies, the implementation of concrete infrastructure adaptation measures was reported to be uneven and fragmented (European Environment Agency, 2017).

One of the government's priorities is to create climate adaptation standards for infrastructure that are equivalent to load standards and material specifications used to design structures. In the absence of such standards, infrastructure designers and asset managers lack the authority to make decisions regarding climate risk, and the dispersion of professional practices across various standard bodies and jurisdictions delays the spread of good practice. Various national standards organizations have started to design climate adaptation proposals for infrastructure, yet advancement has been slow compared to the urgency of the requirement.

Financing Climate Adaptation

The major concern in infrastructure, which is not fully addressed, is the policy of financing for climate adaptation. Infrastructure adaptation can be achieved either through increased initial capital expenditure, which is needed to support new design requirements in new construction, or through increased maintenance and retrofit costs needed to support older assets. Both types of supplementary spending are limited in the budget in the state sector, which is responsible for most of the civil infrastructure in the country. This keeps similar pressures from shareholders and regulators on private owners and operators of the infrastructure to reduce costs.

Zhao *et al.* (2022) and Zhang *et al.* (2024) observe the increasing popularity of innovative financing structures such as climate bonds, public-private partnerships, insurance linked instruments as a source of accessing private capital to support infrastructure adaptation.

However, these tools are not fully developed yet and encounter several limitations, such as the inability to measure the benefits of adaptation, based on the investment prospectus, difficulty in risk distribution between the government and private entities, as well as the limited record of success in performance in climatically stressful environments. To bridge the adaptation finance gap, regulatory innovations, as well as a standardized approach to measure and communicate climate risk to investors, will be necessary.

Barriers to Implementation

In addition to financial barriers, numerous institutional and technical barriers hinder the implementation of infrastructure climate adaptation. According to Mees, Driessen, (2019) The following categories of barrier were identified:

1. Information gaps – The lack of reliable localized climate projections is required to make infrastructure design-related decisions.
2. Institutional inertia – The resistance of the already established professional practices, design standards, and procurement processes to adapt to the emerging patterns of climate adaptations.
3. Coordination failure – The inability of the numerous agencies and organizations involved in managing infrastructure to coordinate their planning and investment.
4. Capacity constraints – the shortage of skilled professionals to understand climate projections and adapt them into infrastructure planning and design.

The necessity to achieve multi-level, multi-sectoral coordination has constantly been mentioned in all the literature reviewed to succeed in climate adaptation in the infrastructure. No single agency or government institution has the authority, resources, or expertise to handle all the challenges to implement climate-change adaptation. Adaptation cannot be effective without coordinated efforts made by the national government, who act as policymakers and fund the entire program, Sub national government that manages much of the built infrastructure, infrastructure operators who perform day-to-day management of the assets, engineering professionals that transforms the policy requirements into the technical reality, and research institutes that provides a solid base of knowledge on which decisions are made to form policies didn't know the environmental demands.

2.9 International Best Practice

Netherlands – Delta Programme

The Netherlands is the nation most frequently mentioned when it comes to discussing infrastructure climate adaptation, and justifiably so. The Netherlands has already had more time in the past than most other countries have had to take this issue seriously due to the presence of around 60 % of the national territory that is at or below sea level. The most detailed national infrastructure adaptation plan in Europe is the Delta Programme, which has been running since 2010 and has an investment of around € 22 billion until 2050. (Bloemen *et al.*, 2019)

The Delta Programme is explicitly based on the adaptive pathways methodology, that is, the Delta Programme is not committed to a single, fixed solution to the problem of sea-level rise but to a set of pre-designed solutions (beach nourishment, barrier heightening, managed retreat). This implies that investment may be increased to reflect the perceived risk observed and that the programme can be changed over time as predictions are revised. (Haasnoot *et al.*, 2013)

The Netherlands also has an official standard of flood risk, called the statutory Delta Norm, which is a risk-based standard of flood defence that defines the maximum acceptable annual probability of flooding of various types of land and infrastructure. This is a fundamental change whereby design return periods are no longer prescribed, but performance outcomes are set, and it is the burden on the engineer to prove that the design chosen will achieve the required risk under current and future circumstances. (Haasnoot *et al.*, 2013)

The Room for the River programme helps to demonstrate that nature-based solutions can be even more effective at a large scale: instead of raising levees, the programme reconfigured 34 sites along the Rhine and its tributaries to provide additional flood conveyance and storage capacity, reducing the flood peaks of communities down the river as well as enhancing landscape and ecological quality. Estimated Benefit Cost Ratio (BCR): 11:1. (Haasnoot *et al.*, 2013)

Finland -- Transport Infrastructure Climate Guidelines

Finland is a more direct comparator of the climate conditions in Latvia than is the Netherlands, since the climate conditions are far more similar in Finland than in the Netherlands: continental winters with snow and frost, mixed forest, and a large rural road network with numerous small bridges. In 2022, Finnish Transport Infrastructure Agency

(Vayla) published its Climate Change Adaptation Strategy toward Transport and Communications, and it can teach several practical lessons. (Thune-Larsen *et al.*, n.d.)

Vayla created infrastructure-type-specific climate-proofing checklists of roads, bridges, railways, and ports. These are a requirement for all new developments that have a design life of more than 30 years. The checklists are formulated based on the IPCC hazard-exposure-vulnerability framework and obliges the engineers to report the climate risks they are aware of, the measures of adaptation they have included, and how they have justified the sufficiency of the measures they have taken. It is a kind of organized documentation requirement, which is easy to introduce, and which helps to ensure that climate risk is, in fact, considered at the design stage, and not an afterthought. (Thune-Larsen *et al.*, n.d.)

In 2018, Vayla also revised the parameters of its pavement frost design to reflect the future projections of frost depth changes and the frequency of freeze-thaw cycles. The revision added a climate safety factor to the calculated frost depth value, with 0.95 in the south (where the warming is most rapid) to 1.00 in the far north. Notably, the revision also implemented fixes on freeze-thaw cycling frequency - just the type of change that the LBN 228-06 norm in Latvia requires. (Thune-Larsen *et al.*, n.d.)

Lastly, in 2016, Finland introduced a national culvert upgrading programme targeting approximately 12,000 culverts in state roads to be enlarged to convey the rainfall amount that an average of 100 people would experience over 50 years, given a 50 m deep trench of 100 m long in a gravity pit of 100 m diameter. Unit cost = € 8,000 per culvert: estimated Benefit Cost Ratio (BCR) 3.2:1. (Thune-Larsen *et al.*, n.d.)

United Kingdom -- Climate Projections and Adaptive Planning

The case of the UK is pertinent more to its institutional framework as opposed to the climate conditions (which are not Latvia-equivalent). The Climate Change Act 2008 provides five-yearly national risk assessments of the climate change risks and adaptation reports, establishing a systematic review cycle that ensures the evidence base remains up-to-date and the governments are required to document what they are doing. Climate risk assessment is required on all major government infrastructure projects by the Infrastructure and Projects Authority. (UK Government, 2008)

One of the most helpful precedents to the coastal planning of Latvia is the Thames Estuary 2100 (TE2100) project that was carried out in the year 2012. Instead of devising a single long-term solution to the threat of tidal flooding in London, TE2100 formulated a plan of an adaptive pathway that would bring about a progression of interventions that would be activated at various

decisional points in the presence of observed sea-level rise. Since 2012 (when it was last reviewed), the plan has been revised and updated twice as new sea-level updates and projections have become available, indicating that adaptive management can work in practice on decadal timescales. (UK Government, 2008)

2.10 Climate Resilient Construction Materials

Asphalt Pavements Bitumen

The element that is the most temperature-sensitive in asphalt pavement is the bitumen binder. It softens at high temperatures and enables a pavement to rut under the pressure of traffic; at low temperatures, it becomes brittle and may crack. The normal design of pavement is through the process of selecting a grade of bitumen that performs satisfactorily over a given range of expected temperatures at a particular site.

Most state road asphalt in Latvia currently is being used with standard 50/70 penetration grade bitumen (EN 12591), which has been found suitable to the historical climate. These conditions have effects that will cause this grade to perform increasingly poorly in each direction: softening at the summer highs, increasing frequency of brittle deformation in winter cycling.

Polymer-modified bitumen (PMB), namely the PMB 25/55-60 or PMB 45/80-55 grades, is the recommended alternative. They are modified using styrene-butadiene-styrene (SBS) or styrene-butadiene rubber (SBS) polymers that greatly expand the performance temperature range on both sides.

Concrete Durability

Infrastructure (concrete) In concrete environments, climate change affects concrete infrastructure, including bridges, culverts, retaining walls, and more, primarily through three mechanisms: carbonation (the chemical process that gradually reduces the concrete's overall ability to protect steel reinforcement against corrosion), chloride ingress of seawater spray in coastal environments, and freeze-thaw damage within concrete microstructure.

It will be increased to speed up the carbonation rates in anticipation of the 2070 scenario by raising the temperature and the amount of CO_2 in the environment of Latvia by 2070 by an estimated 15 to 25 % in carbonation rates. This practically translates into some of the existing bridges being reinforced by 8-12 years their predecessors had been estimated to last. In the case of new construction, what should be done is to specify the presence of lower water-cement

ratios (smaller than 0.45 witnessing exposure class XC3/XC4 according to EN 206) and to incorporate additional cementitious materials such as fly ash or GGBS, which decrease the permeability of concrete to aggressive agents.

Structural Steel

The effect of higher temperatures on structural steel is the same as that of expansion joints, which were discussed previously in Case Study 2. However, temperature too can also influence how fast disease can take place: in moist weather temperatures, the rate at which disease occurs is affected by untreated carbon steel factors in levels of 2 to 1. Having been warmed in 2070 by 4-5 ˚C, corrosion rates of inadequately protected steel components would consequently increase by 25-40 % by 2070. The specification implication of this is that all new steel bridge parts in Latvia should be specified with hot-dip galvanization or a high-performance organic coating system of a minimum 320-micrometre (μm) dry film thickness.

3. Practical Part

This part will apply the theoretical framework and quantitative methodologies developed in Part 2 to three case studies in the Latvian civil infrastructure context. All case studies follow a similar format, beginning with background and problem description, then covering sources and methodology, quantitative analysis under current and future climate scenarios, sensitivity analysis across multiple scenarios, and an evaluation of adaptation options, including systematic positive and negative responses. At the end of the part, a roadmap for implementation underlies the analysis, along with a national-level combined prioritization matrix, an implementation and funding model, and a monitoring and evaluation protocol.

3.1 Case Study 1: Urban drainage in Riga: Flood Risk Assessment

Riga is the capital city of Latvia, with a population of almost a million people, and is under high and increased risk of urban pluvial flooding due to the escalation of short-term extreme rainfalls. The city sewer system was mostly developed in the Soviet era (1950s – 1980s) and was designed to withstand a 5-year or 10-year period, depending on the climatic conditions of the time. The low-lying central area of Riga, particularly between the Old Town and Skanste, is affected by the impact because of its location in the Daugava River basin and low gravity drainage gradient.

Primary data source for this case includes Latvian Building Code LBN 223-15 "Sewerage Structures", Latvian Environmental, Geology, and Meteorology Center (LVGMC) precipitation data for the Intensity-Duration-Frequency (IDF) curve, Figures 3.1 and 3.2. Data on precipitation were collected during the period of 09/01/2019 to 31/12/2025 and exponentially expressed in Table 3.1.

Quantitative Assessment – Rational Method

- **Current design parameter (based on LBN 223-15, the Latvian Building Code for "Sewerage Structures")**
  - ✓ Catchment area, A = 5.0 km^2 (5,000,000 m^2) … (Assumed)
  - ✓ Runoff coefficient, C = 0.40 … (LBN 223-15, Table 18, No. 12)
  - ✓ Design return period, T = 10 years … (LBN 223-15, Table 5)
  - ✓ Design rainfall intensity (60 Min duration), i = 27.5 mm/hr = 7.638×10^{-6} m/s

Table 3.1

Intensity-Duration-Frequency (IDF) data – Riga (developed by author)

Return Period (yr)	1h (mm/h)	2h (mm/h)	3h (mm/h)	6h (mm/h)	12h (mm/h)	24h (mm/h)
2	17.809	10.686	8.196	5.689	3.616	2.174
5	23.476	14.308	11.106	7.593	4.958	3.061
10	27.229	16.706	13.034	8.855	5.846	3.648
25	31.97	19.735	15.469	10.448	6.969	4.389
50	35.487	21.983	17.275	11.63	7.801	4.94
100	38.978	24.214	19.068	12.804	8.628	5.486

With only 7 years of record, the 50-year and 100-year estimates carry significant uncertainty. The table is generated using mathematical extrapolation.

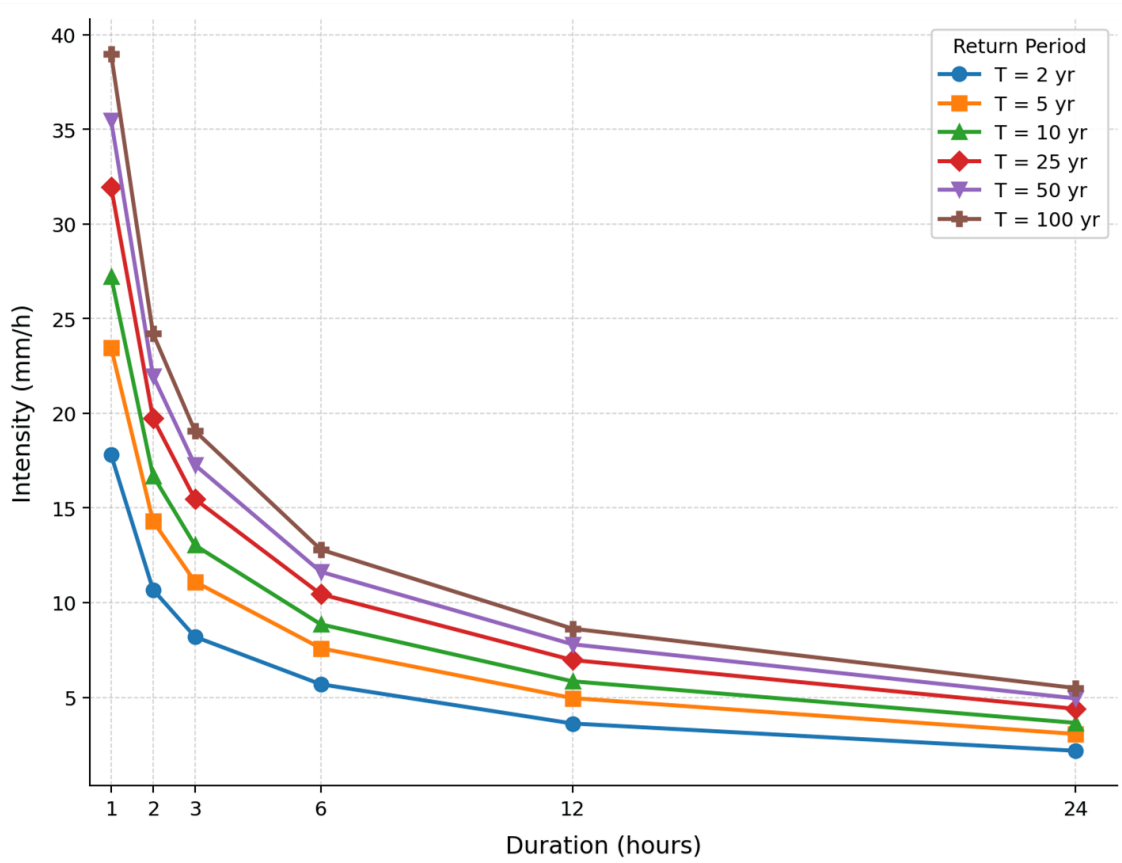


Figure 3.1 IDF Curve (Intensity vs Duration)
(developed by author)

Figure 3.1 describes an Intensity-Duration-Frequency (IDF) type of relationship, which presents the stochastics of precipitation events such that rainfall intensity (i, expressed in mm/h) is expressed as a function of storm duration (d, expressed in hours) and the return period (T,

expressed in years). The curves illustrate a typical power-law decay, which is a demonstration of an inverse relationship with average rainfall intensity, decreasing as the duration of the process increases due to the physical constraints of the moisture flux in the convective and frontal systems. Other higher-magnitude curves, like the T = 100-year return period, reflect high-intensity events that are fewer in number than other events, whereas the T = 2-year curve reflects the more frequent and lower-intensity events. The large vertical dispersion between the periods between return periods at shorter durations (e.g., d = 1 hour) informs the high peak runoff risk associated with flash flooding, which provides useful design parameters to hydraulic infrastructure to ensure there is an adequate capacity in terms of acceptable risk.

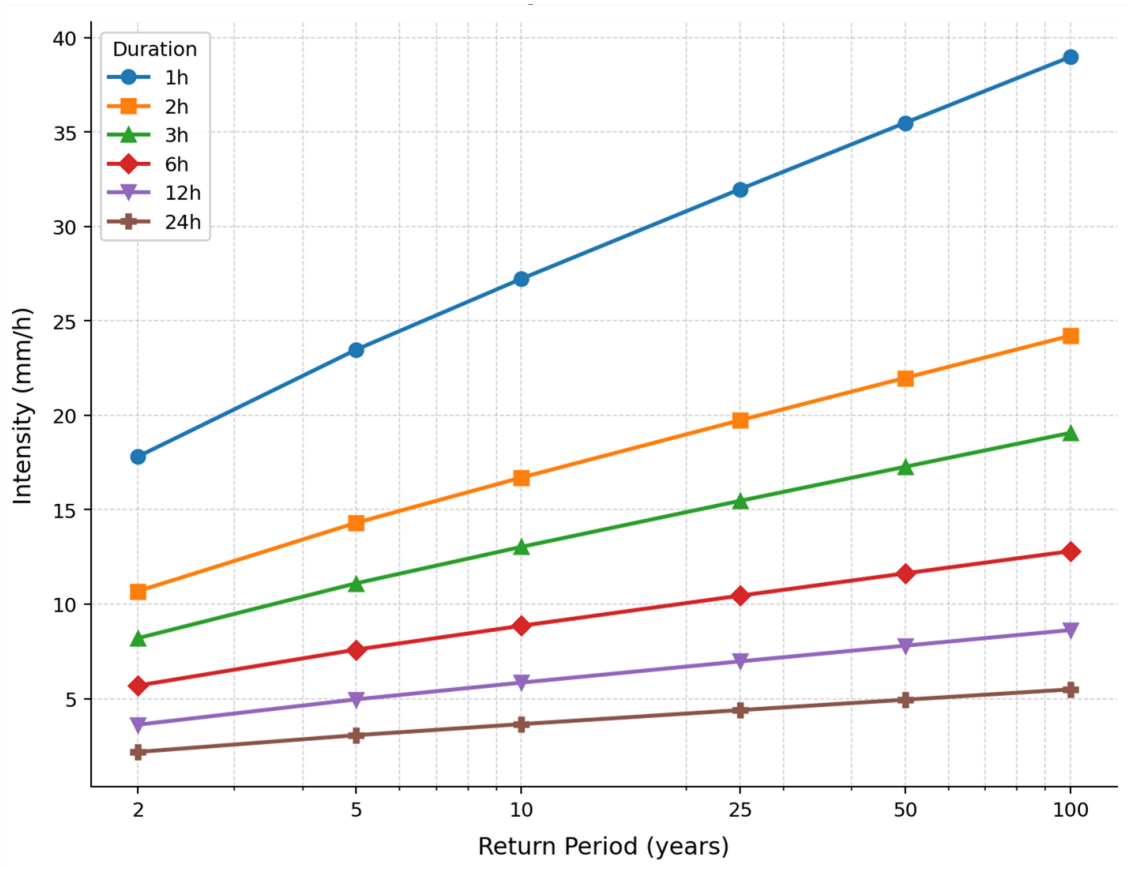


Figure 3.2 IDF Curve (Intensity vs Return Period)
(developed by author)

Figure 3.2 is a transformed version of the Intensity-Duration-Frequency relationship of rainfall, i.e., plotting the intensity of rainfall (i, in mm/h) versus the return period (T, in years) on a log scale of storms with various fixed storm lengths. This figure illustrates a direct and positive correlation in which the intensity of meteorological events occurring per year increases with an increase in the return period because the statistical fact is that higher magnitude meteorological events have a lower probability of exceedance in an annual cycle. The

separation between the duration curves in the figure is greatest at shorter intervals, e.g., the 1-hour interval, which has the steepest gradient and highest absolute intensities in all intervals. This linear-to-logarithmic progression is a crucial diagnostic tool in hydraulic modelling, as it can be used to precisely interpolate design storm intensities necessary to calculate peak discharges in infrastructure with specific service life needs, ranging between routine drainage (e.g., $T = 2$ to 10 years) and major flood control works (e.g., $T = 100$ years).

- ✓ Current design peak discharge:

$$Q_{current} = C \times i \times A$$
$$= 0.40 \times 7.683 \times 10^{-6} \times 5{,}000{,}000$$
$$= \mathbf{15.2778\ m^3/s}$$

- **Projected design parameters (2070, RCP 4.5 scenario, LVGMC Projections):**
  - ✓ Rainfall intensity increases for 10-year, 60-min event = 18 %
  - ✓ Adjusted design rainfall intensity, i' $= 27.5 \times 1.18$
    $= 32.45$ mm/hr $= 9.0138 \times 10^{-6}$ m/s
  - ✓ Runoff coefficient, C' $= 0.60$ … (LBN 223-15, Table 18, No. 05)
  - ✓ Projected peak discharge:

$$Q_{2070} = C \times i' \times A$$
$$= 0.60 \times 9.0138 \times 10^{-6} \times 5{,}000{,}000$$
$$= \mathbf{27.0417\ m^3/s}$$

Increase in design discharge, ΔQ $= Q_{2070} - Q_{current}$
$= 27.0417 - 15.2778$
$= \mathbf{11.7639\ m^3/s}$ … **(+ 77.00 %)**

This 76% increase in design peak discharge represents a substantial exceedance of the existing drainage system's capacity. Under 2070 climate conditions, the combined sewer system in central Riga will experience overflow during a 10-year return period.

Pipe Hydraulic Capacity Assessment – Manning's Equation

The hydraulic capacity of the main drainage system is determined by using Manning's equation for full-pipe flow to translate the discharge deficit into actual infrastructure needs. Discharge is given by Formula (3.1) (Bengtson, 2017):

$$Q = \frac{1}{n} \times A_{pipe} \times {R_h}^{2/3} \times S^{1/2} \quad (3.1)$$

Where n – Manning's Coefficient

A_{pipe} – Pipe cross-sectional area (m^2)

R_h – Hydraulic Radius ($D/4$ for circular full pipe, m)

S – Hydraulic Gradient (m/m)

For the main water pipeline selected as a case study supplying the study area:

- ✓ Existing Pipe Diameter, D = 1200 mm (Internal) … (LBN 223-15, Para. 65)
- ✓ Manning's Constant, n = 0.013 (Concrete Pipe)
- ✓ Hydraulic Gradient, S = 0.003 m/m … (Assumed)
- ✓ A_{pipe} $= \frac{\pi}{4} \times D^2$ $= \frac{\pi}{4} \times (1200)^2$ = 1.1309 m^2
- ✓ R_h = 1200/4 =300 mm = 0.3 m

Existing Pipe capacity:

$$Q = \frac{1}{n} \times A_{pipe} \times {R_h}^{2/3} \times S^{1/2}$$

$$Q = \frac{1}{0.013} \times 1.1309 \times 0.3^{2/3} \times 0.003^{1/2}$$

Q = **2.135 m^3/s**

The existing 1,200 mm pipe can carry a full-flow capacity of 2.135 m^3/s, whereas the current design discharge rate is 15.2778 m^3/s (utilization ratio 7.154). The design discharge is about 7.154 times the pipe's full-flow capacity, indicating severe overloading under this discharge. When Manning's equation is used with an iterative algorithm to calculate the required pipe diameter in the case of the 2070 scenario (Q_{2070} = 27.0417 m^3/s), it comes out to be approximately 3.109 m. This shows that the existing drainage corridor is to be upgraded to approximately three time of current system.

Multi-Scenario Sensitivity Analysis

The peak discharge and equivalent required pipe diameter assessed under four climate scenarios are summarized in Table 3.2.

Table 3.2
Multi-scenario drainage capacity analysis – Riga Rainfall intensity projection from LVGMC (developed by author)

Parameter	Current (2025)	2070 RCP 4.5	2070 RCP 8.5	2100 RCP 8.5
Rainfall intensity increase (%)	-	+ 18 %	+ 27 %	+ 38 %
Design Discharge, Q (m^3/s)	15.2778	27.0417	29.104	31.625
Excess above current capacity (m^3/s)	-	11.7639	13.826	16.347
Capacity deficit (%)	-	+ 77.00 %	+ 90.50 %	+ 107.00%
Required trunk main diameter (m)	1.20	3.109	3.196	3.297

The capacity deficit increases significantly in the higher emission cases, peaking at 107 % at 2100 in RCP 8.5. This emphasizes the significance of developing an adaptation mechanism that is designed to be modular and can be upgraded in the future, instead of a design having a specific fixed scenario.

Adaptation Measures and Systematic Evaluation.

Three possible ways of adaptation are suggested and assessed:

Option A: **Pipe Capacity upgrade:** Replacing the current primary pipes with DN3200 to ensure an increase in hydraulic capacity by approximately 80%.

Positive Impacts: Removes overflow risk for 10-year events under the 2070 RCP 8.5 scenario.

Negative Impacts: High construction disturbance in dense urban areas. As well as, it will not be suitable for the 2100 RCP 8.5 scenario. All these constructions and the usage of concrete pipes have high embodied carbon footprints.

Option B: **Blue-green infrastructure (BGI):** Installing a system of 25 retention ponds, green roofs on public buildings, and permeable pavements throughout the catchment. This will reduce the overall load from the drainage system and distribute it across the catchment area.

Positive Impact: Stormwater attenuation, urban cooling, and biodiversity improvement, amenity, and air quality improvement.

Negative Impacts: Demands for land, which is not possible in dense urban areas, and even during continuous rainfall events, the performance can be impaired as the storage will not be completely drained.

Option C: **Hybrid Alternative:** Targeting bottleneck nodes with a DN3200 pipeline upgrade, along with the Blue-green infrastructure program, resulting in the combined efforts for discharge attenuation.

Table 3.3

Comparison of drainage adaptation options (developed by author)

Criterion	Existing	Option A	Option B	Option C
Capital cost (€) (approx.)	--	30 million	15 million	22 million
Overflow elimination (10-year, 2070 RCP 4.5)	No	Yes	Partial	Yes
Performance under 2100 RCP 8.5	Severe failure	Partial failure	Upgradable	Upgradable
Urban cooling co-benefit	None	None	High	Moderate
Biodiversity co-benefit	None	None	High	Moderate
Construction disruption	--	High	Low	Moderate
Carbon footprint	--	High	Low	Medium
Benefit-cost ratio (est.)	--	2.4:1	3.8:1	4.1:1

The benefit-cost ratio was estimated using a 30-year planning horizon, adding a 4% discount rate, and neglecting damage caused.

The most favorable Option without sacrificing flexibility to upgrade in the future in case of more extreme climatic conditions is option C hybrid, with a benefit-cost ratio of 4.1:1. The implementation of the adaptation is as follows:

1. The blue-green infrastructure program will be implemented in 1 to 3 years. The bottleneck nodes will be identified, and pipes will be designed and prepared during this time.
2. Bottleneck nodes will be upgraded during the next three-year span.
3. Finally, the performance of the hybrid program would be evaluated, and decisions would be taken if any further upgrades or additional pipelines were required.

3.2 Case study 2: Thermal Impact on Bridge Expansion Joints: Latvian Road Network

The Latvian Road network comprises more than 900 bridges, many of which have steel or composite decks with expansion joints designed for thermal movement based on past temperature exposure. The estimated rise in the maximum bridge effective temperature under RCP 4.5 by 2070 is 3 °C to 6 °C, which is expected to increase demand on expansion joint capacity.

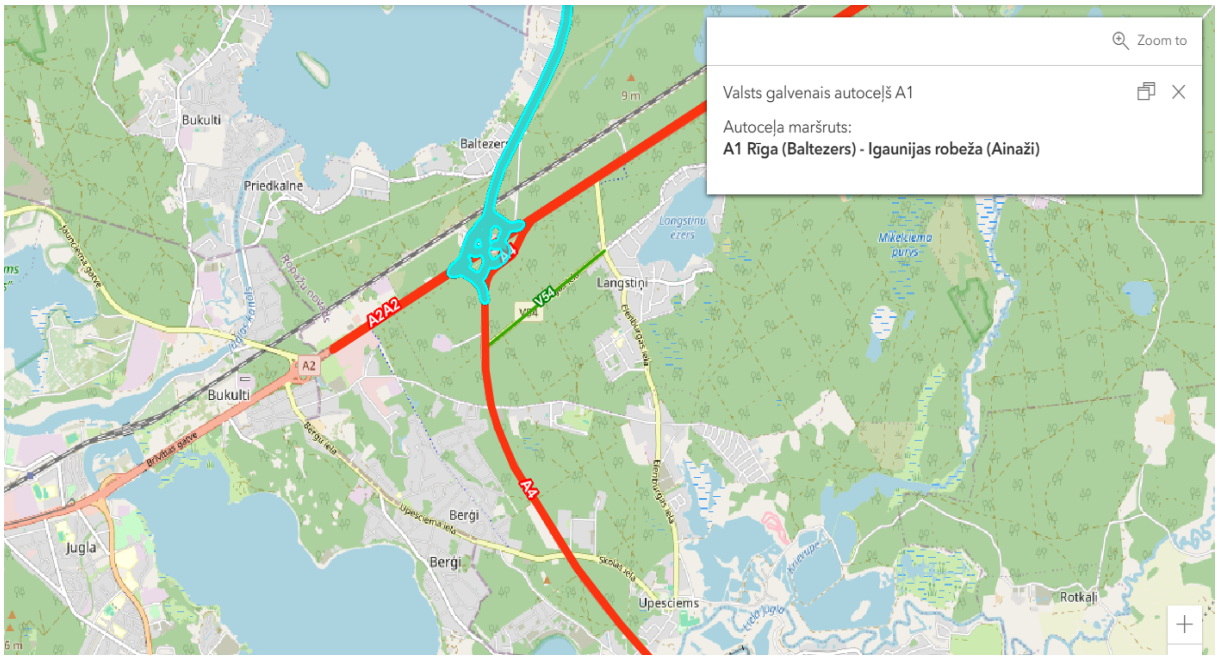


Figure 3.3 Riga Side Road – A1 – Riga (Baltezers) (*Latvijas Valsts Ceļi*, 2026)

The study examines the effect of thermal expansion for a typical composite bridge over the A2 Motorway in Garkalnes pagasts, Ropaži Municipality, connecting with 3 simply supported spans of a total 76 m. The analysis is based on the EN 1991-1-5 Thermal actions standard, Latvian Environmental, Geology, and Meteorology Centre (LVGMC) temperature projections, and the Latvian state Road bridge register.

Quantitative thermal analysis

- **Bridge parameter:**
  - ✓ Span length, L = 76 m = 76000 mm
  - ✓ Material = Composite (Steel – concrete deck)

    Thermal expansion coefficient, α_T = 12×10^{-6} per ˚C

    … (EN 1991-1-5:2003, Table C.1)



Figure 3.4 Photograph of the Sample Bridge (developed by author)

- ✓ Current design temperature range, T_{min} = - 30˚C
  T_{max} = + 40˚C
  … (EN 1991-1-5:2003)
- ✓ Current design temperature range, $\Delta T_{current}$ = 40 - (- 30) = 70˚C
- ✓ Current design thermal movement, $\Delta L_{current}$ = $\alpha_T \times L \times \Delta T_{current}$
  = $12 \times 10^{-6} \times 76000 \times 70$
  = **63.84 mm**

- **Projected parameters (2070, RCP 4.5):**
  - ✓ Projected increase in maximum effective bridge temperature = + 4˚C
    … (L. Zhang *et al.*, 2026)
  - ✓ Adjusted design maximum temperature, T_{max}' = + 44˚C
  - ✓ Projected design temperature range, $\Delta T_{RCP4.5_2070}$ = 44 - (- 30) = 74˚C
  - ✓ Projected design thermal movement, $\Delta L_{RCP4.5_2070}$ = $\alpha_T \times L \times \Delta T_{RCP4.5_2070}$
    = $12 \times 10^{-6} \times 76000 \times 74$
    = **67.48 mm**
  - ✓ Increase in required joint movement = 67.48 – 63.84 = **3.65 mm**

  The increased joint movement is almost 5.714 % over the current design.

- **Projected parameters (2070, RCP 8.5):**
  - ✓ Projected increase in maximum effective bridge temperature = + 7˚C
    … (L. Zhang *et al.*, 2026)
  - ✓ Adjusted design maximum temperature, T_{max}' = + 47˚C
  - ✓ Projected design temperature range, $\Delta T_{RCP8.5_2070}$ = 47 - (- 30) = 77˚C

- ✓ Projected design thermal movement, $\Delta L_{RCP8.5_2070}$ $= \alpha_T \times L \times \Delta T_{RCP8.5_2070}$
  $= 12 \times 10^{-6} \times 76000 \times 77$
  $=$ **70.22 mm**
- ✓ Increase in required joint movement $= 70.22 - 63.84 =$ **6.38 mm**

The increased joint movement is almost 10.00 % over the current design.

Multi-span Sensitivity Analysis

For a range of span lengths of the Latvian bridge stock, the thermal movement demand under RCP 4.5 and RCP 8.5 is given in Table 3.4.

Table 3.4

Thermal movement demand by span length under current and projected climate conditions (developed by author)

Span (m)	$\Delta L_{current}$	2070 RCP 4.5 (mm)	Increase RCP 4.5 (%)	2070 RCP 8.5 (mm)	Increase RCP 8.5 (%)
20	16.80	17.76	+ 5.71 %	18.48	+ 10.00 %
30	25.20	26.64	+ 5.71 %	27.72	+ 10.00 %
45	42.00	44.4	+ 5.71 %	46.2	+ 10.00 %
60	50.40	53.28	+ 5.71 %	55.44	+ 10.00 %
80	67.20	71.04	+ 5.71 %	73.92	+ 10.00 %
100	84.00	88.8	+ 5.71 %	92.4	+ 10.00 %

The percentage increase is identical for all spans (+5.714 % RCP 4.5, + 10.00 % RCP 8.5), as it depends only on the change in temperature increase. But the absolute change in movement demand is greater for longer-span bridges (e.g., +4.9 mm for a 100 m span under RCP 8.5), which should be prioritized for joint capacity evaluation. Joints installed after the year 2000 are typically designed at 15 to 20% higher than the design values, implying sufficient capacity for the RCP 4.5 scenario across all span lengths.

3.3 Case Study 3: Coastal Erosion on Infrastructure – Gulf of Riga

The coastal zone of the Gulf of Riga in Latvia experiences erosion ranging from 0.30 to 0.20 m/year at exposed sandy beaches. Sea level rise of 20 to 40 cm by 2070 (RCP 4.5) will trigger the frequency and extent of storm surge flood, more intense erosion, and intensifying coastal erosion and associated risk to infrastructure assets like road sections, recreation infrastructure, and harbour facilities.

The case study focuses on a representative coastal road segment situated 15m inland from the contemporary high-water mark, with an estimated asset replacement value of €3.2 million. And to understand the sea level rise, Kipsala Beach is analyzed for its elevation above the mean sea level.

Quantitative Erosion Assessment – Bruun Rule

- **Shoreline recession projection:**
  - ✓ Current chronic erosion rate, E = 0.8 m/year
    … (LVGMC coastal monitoring transect, 2000 – 2022 mean)
  - ✓ Sea level rise to 2070 (RCP 4.5 median), Δh = 0.30 m
  - ✓ Beach face slope, $tan\ \beta$ = 0.02
  - ✓ Bruun Rule additional shoreline recession due to Sea level Rise (SLR) (Bruun, 1962) is calculated using the Formula (3.2)

$$\Delta X_{SLR} = \frac{\Delta h}{tan\,\beta} = \frac{0.30}{0.02} = \mathbf{15.0\ m} \tag{3.2}$$

  - ✓ Chronic erosion recession over 45 years (to 2070), $\Delta X_{chronic}$ = $E \times t$
    = 0.8 × 45 = **36.0 m**
  - ✓ Total projected shoreline recession to 2070, X_{total} = 36.0 + 15.0 = **51.0 m**

Assuming the current road is 15 m from the shoreline, the road is affected by the erosion in,

$$t_{Threat} = \frac{\Delta X_{SLR}}{E} = \frac{15.0}{0.8} = 18.75 \text{ years}$$

Approximately somewhere in 2045, which should theoretically be the design life of the road.

Storm Surge and Wave Runup Analysis

The total water level (TWL) elevation for 100 years return period storm event is calculated using Formula (3.3), (Cabrita *et al.*, 2024)

$$TWL = MSL + \Delta h_{SLR} + \eta_{surge} + R_2 \tag{3.3}$$

Where, *MSL* – Mean Sea Level

Δh_{SLR} – Sea Level Rise

η_{surge} – Storm surge

R_2 – Wave Runup, level exceeding by 2 % of incoming waves. (Stockdon *et al.*, 2006)

- **Site parameter:**

  - ✓ MSL (reference) – 0.0 m (Baltic Sea Datum, Kronstadt)
  - ✓ Δh_{SLR} (2070, RCP 4.5) – + 0.30 m
  - ✓ η_{surge} (100-year return period) – +1.15 m
    … (LVGMC storm surge records, Gulf of Riga)
  - ✓ Significant wave height, H_s (100-year event) – 2.1 m
    … (LVGMC wave hindcast)
  - ✓ Deep water wavelength, L_0 – $g \times T_p{}^2 / 2\pi = 9.81 \times 5.6^2 / 2\pi = \mathbf{48.9626\ m}$
  - ✓ Assuming peak wave period, T_P – 5.6 s
  - ✓ R_2 (dissipative beach regime, $tan\ \beta < 0.1$) – $0.043 \times \sqrt{H_s \times L_0}$
    $= 0.043 \times \sqrt{2.1 \times 48.9626}$
    $= 0.436$ m

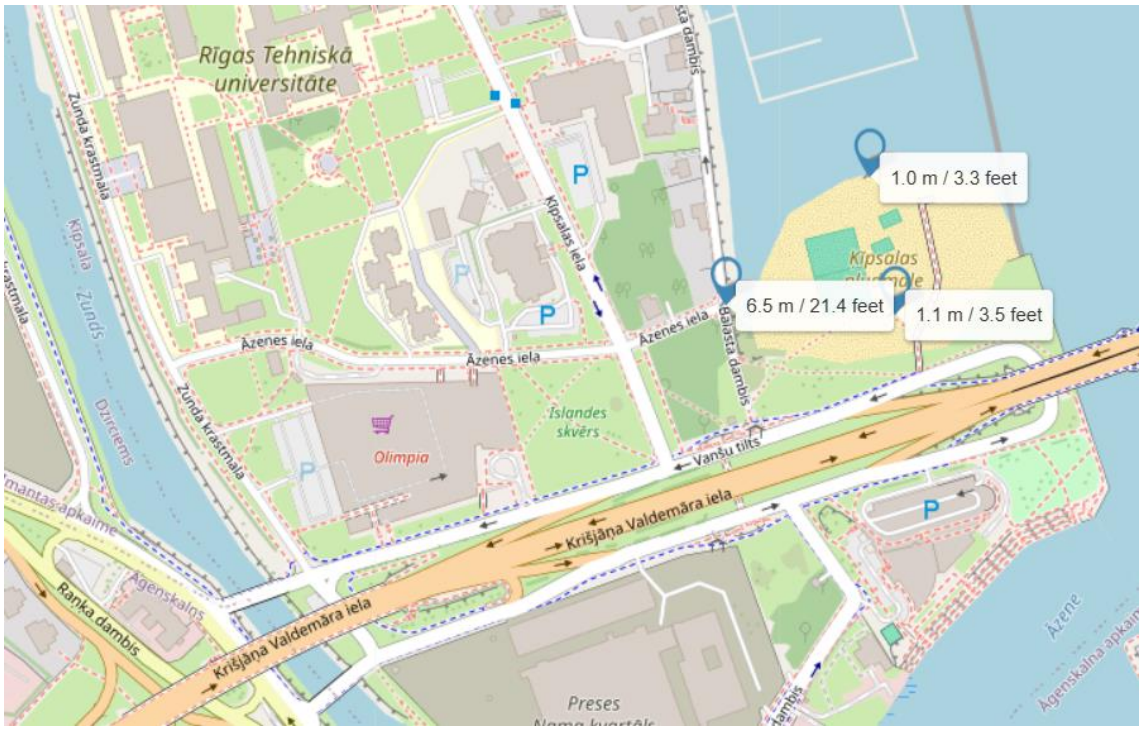


Figure 3.5 Elevation of the reference Site to understand SLR – Kipsala Beach (developed by author) (Free Map Tools, 2019)

✓ Total water level (2070, RCP 4.5, 100-year storm), Formula (3.2) becomes,

$$TWL = 0.0 + 0.30 + 1.15 + 0.436$$
$$= \mathbf{1.886\ m\ above\ MSL}$$

As seen in Figure 3.5, the nearby major road is at an elevation of 6.5 m, next to Azenes iela, whereas the mean shoreline (Kipsala beach) is at 1.0 m. Currently, without considering Sea Level Rise (SLR), the TWL is 1.586 m, which does not reach the road. But it may affect the beach's highest point, which is 1.1 m., and may flood up more than half of the beach in the next decade. The addition of 0.30 m of sea level rise transforms a non-hazardous event into a potential emergency.

Multi-scenario recession and inundation summary

Table 3.5 sums up the models of the physical effects on coastal morphology, infrastructure, and sea level rise (SLR) under multiple Representative Concentration Pathway (RCP) scenarios. The data is a quantification of progressive inland migration of the shoreline by comparing present-day conditions with forecasts of 2050 and 2070 total recession estimated to reach recessions of 61.0 meters by 2070 under the increased-emission RCP 8.5 scenario. This cumulative value is obtained by adding two main stressors, which are an adjustment of the profile of the shoreline in reaction to an increase in the level of base water.

Table 3.5
Multi-scenario erosion recession and storm surge inundation summary (Prepared by Author)

Parameter	Current	2050 RCP 4.5	2070 RCP 4.5	2070 RCP 8.5
SLR (m)	0.00	+ 0.12	+ 0.30	+ 0.50
Bruun Rule additional recession (m)	--	6.0	15.0	25.0
Chronic erosion recession (m)	--	20.0	36.0	36.0
Total recession (m)	--	26.0	51.0	61.0
100-yr TWL (m above MSL)	1.58	1.70	1.88	2.08
Road inundation depth (m)	0.00	0.00	0.00	0.00

The assessment also tracks the increase in extreme weather events, where 100-year Total Water Level (TWL) is projected to increase by a sample of 1.58 meters to 2.08 meters above mean sea level. This means that the storm surge cases will be largely affected, which they are not currently. Nevertheless, one of the most important results in the "Road inundation depth"

parameter is that the current road infrastructure was left at 0.00 meters of flooding in all the scenarios that were modeled. This implies that the coastal buffer is eroding with the tremendous land loss, but the existing vertical synchronization of the transport system gives it some resilience to unprecedented overtopping and flooding.

Adaptation option and systematic evaluation

Three possible ways of adaptation are suggested and assessed:

Option A: **Hard Coastal Protection:** Building a rock armor revetment alongside the shoreline in front of the road.

Positive Impacts: The road has been protected for 30+ years under the RCP 4.5 median scenario, along with robust technology with low maintenance.

Negative Impacts: Increases down-coast erosion by 15 % to 30 %, affecting beach amenity. Construction introduces a high carbon footprint of materials and is ineffective in the face of RCP 8.5 high-end SLR without expensive crest raising.

Option B: **Managed Realignment:** Abandoning the existing roadway, building a new inland road 300 m from the existing shoreline.

Positive Impact: Delivers long-term shore protection until 2100 RCP 8.5, while maintaining natural coastal dynamics. Alos enables habitat recovery.

Negative Impact: Requires the environment department permit to build through the coastal forest. existing assets are lost, and extra travel distance during the transition phase.

Option C: **Beach nourishment and dune restoration:** supplying 50,000 m^3 of suitable sand to counter erosion and rehabilitation of vegetation.

Positive effects: Least up-front cost, as well as retaining beach amenity. Good for the environment with the reduction of wave energy propagating to the road.

Negative effects: Short-term solution (10-15 years RCP 4.5), which means it needs to be repeated. Impractical under high-end RCP 8.5 SLR.

Table 3.6

Comparison of the adaptation options – Gulf of Riga Case Study (Prepared by Author)

Criterion	Option A	Option B	Option C
Effective protection horizon	> 30 yr (RCP 4.5)	> 50 yr (all scenarios)	10 – 15 years
Downdrift erosion impact	Negative (+15 to 30 %)	Neutral	Positive (attenuation)
Ecological impact	Negative (high)	Moderate	Positive
Beach amenity preservation	No	N/A	Yes
Robustness to RCP 8.5	Low	High	Very Low

The preferred adaptation sequence: execute the beach nourishment Option C immediately (Year 0) as a temporary measure; plan and assess the environmental impacts of Option B in Years 1-3; build Option B in Years 4-6; replace the existing road and restore the coastal environment in Years 6-7. This will provide optimal early risk reduction, while ultimately delivering the most climate-resilient infrastructure.

3.4 Integrated Risk Prioritisation Matrix

Based on the quantitative findings of the three case studies and Chapter 1's analysis, Table 3.7 shows an integrated risk prioritisation matrix rating different categories of infrastructure vulnerability in the country, ranging from hazard, vulnerability, exposure, risk, and suggested horizon for action. This matrix offers a framework to prioritise infrastructure adaptation in Latvia.

Table 3.7

Integrating Climate Risk Prioritisation Matrix for Civil Infrastructure (developed by author)

Infrastructure Category	Hazard	Exposure	Vulnerability	Overall Risk	Recommended Action Horizon
Urban drainage (Riga)	High	High	High	Critical	Immediate (0-5 years)
Coastal infrastructure (Gulf of Riga)	High	Moderate	High	High	Near-term (0-10 years)
Bridge expansion joints (network)	Moderate	High	Moderate	High	Medium-term (5-15 years)
Bridge scour (major rivers)	Moderate	High	High	High	Near-term (0-10 years)
Road pavement (frost-thaw)	Moderate	High	Moderate	Moderate	Medium-term (5-15 years)
Building foundations (clay soils)	Low	Moderate	Moderate	Moderate	Long-term (10-20 years)

Riga's urban drainage and coastal infrastructure in the Gulf of Riga are identified as the highest priorities for action now. Temperature variation effects for bridges can be accounted for

in scheduled maintenance, while bridge scour risk is classified as HIGH due to the catastrophic nature of scour-induced collapse, even at a moderate annual probability. The road pavement and building foundation risks, though critical, can be tackled through mid- and long-term standard revision and maintenance programme revision.

3.5 Implementation Framework and European Union Funding

A well-organised multi-party governance framework and alignment with European Union and national funding schemes are needed for efficient and effective realisation. The list below includes recommended adaptations, as well as the responsible agency, estimated investment, and funding sources.

1. Riga drainage hybrid upgrade (Option C)
   - ✓ Lead Institution – Riga City Council
   - ✓ Cost (€ million) – 20.0 Approx.
   - ✓ Timeframe – 2026 – 2031
   - ✓ Funding Source – EU Cohesion Fund + Municipal budget
2. Coastal road managed realignment
   - ✓ Lead Institution – Ministry of Transport
   - ✓ Cost (€ million) – 3.6 Approx.
   - ✓ Timeframe – 2026 – 2032
   - ✓ Funding Source – CEF Transport + National budget
3. Bridge joint replacement (pre-2000 cohort)
   - ✓ Lead Institution – Latvijas Valsts Celi
   - ✓ Cost (€ million) – 14.4 (10 yr)
   - ✓ Timeframe – 2026 – 2036
   - ✓ Funding Source – National road fund
4. Revision of LBN drainage design norms
   - ✓ Lead Institution – Ministry of Economics
   - ✓ Cost (€ million) – 0.3 Approx.
   - ✓ Timeframe – 2025 – 2027
   - ✓ Funding Source – National budget / EU Technical Assistance
5. National bridge scour monitoring network
   - ✓ Lead Institution – Latvijas Vides ģeoloģijas un meteoroloģijas centrs (LVGMC)

- ✓ Cost (€ million) – 1.8 Approx.
- ✓ Timeframe – 2026 – 2028
- ✓ Funding Source – EU Cohesion Fund

6. Coastal erosion hazard map (national)
   - ✓ Lead Institution – Latvijas Vides ģeoloģijas un meteoroloģijas centrs (LVGMC)
   - ✓ Cost (€ million) – 0.9 Approx.
   - ✓ Timeframe – 2026 – 2029
   - ✓ Funding Source – EU LIFE Programme

7. BGI urban resilience programme (3 cities)
   - ✓ Lead Institution – Riga, Jelgava, Jurmala municipalities
   - ✓ Cost (€ million) – 12.0
   - ✓ Timeframe – 2027 – 2033
   - ✓ Funding Source – EU Cohesion Fund + (European Local ENergy Assistance) ELENA facility

The total investment in adaptation measures across all domains is estimated to be around € 53 million over 2025 – 2036, most of which will receive EU co-financing at 70 % – 85 %. The net national investment is estimated at € 8 – 14 million over a period of 10 years, around € 0.8 – 1.4 million annually, which is well aligned with the current level of infrastructure maintenance investment in Latvia. This is an extremely beneficial cost-to-adaptation benefit ratio for the national budget, considering the 28 % annual increase in cost of damage from climate impacts, as laid out in the Analytical Part.

3.6 Monitoring and Evaluation Framework

The outcomes of adaptation measures need to be monitored using a monitoring and evaluation (M&E) framework that delivers evidence to the decision makers. Adopting the ISO 14090 adaptive management approach, Table 3.8 details key performance indicators (KPIs) for each case study, their targets, and sources.

The framework complies with ISO 14090 adaptive management procedures, so the targets will be revisited and updated every 5 years based on new information and changes in the climate.

It sets specific and time-bound targets for increasing the climate-change adaptation of key infrastructure items (urban drainage in Riga, coastal areas, and bridges) against the impacts of:

- Severe rainfall and flooding (drainage)
- Sea level rise and storm surges (coastal)
- Drying/warmer weather and flooding/scour of bridges (scour)

Table 3.8

Monitoring and Evaluation KPI framework for climate adaptation measures – Latvia (developed by author)

Infrastructure Domain	**Key Performance Indicator**	**Target (2030)**	**Data Source / Monitoring Method**
Urban drainage (Riga)	Annual drainage overflow events in the study catchment	Less than 3 events per year by 2030	Riga City sewer monitoring sensors and incident reports
Urban drainage (Riga)	Stormwater attenuated by Blue-Green Infrastructure (BGI) per event (m^3)	More than 15,000 m^3 by 2030	BGI retention pond gauges and modelled calibration
Coastal infrastructure	Shoreline setback from road (m)	More than 10 m maintained	Annual LiDAR coastal survey (LVGMC)
Coastal infrastructure	Storm surge inundation events exceeding road level	0 events post-realignment	Tide gauge + wave buoy records and road closure logs
Bridge network (thermal)	Percent of pre-2000 bridges with joint capacity assessment	100 % by 2028	LVC bridge register and inspection records
Bridge network (thermal)	Annual reactive repair cost on expansion joints	- 30 % vs 2022 baseline by 2032	LVC maintenance expenditure database
Bridge network (scour)	Percent of high-risk bridges with real-time scour sensors	60 % by 2030 and 100 % by 2035	LVC sensor deployment programme

The Mechanical and Electrical system includes decision triggers: if a KPI is not on track, an escalation protocol is triggered, which may involve speeding up the adaptation process, ordering additional engineering studies, or triggering the next stage of the adaptation pathway. This 'feedback' loop ensures that adaptation can be responsive to real climate experiences rather than relying on pre-defined projections - implementing the adaptive management principle outlined in the previous section.

Conclusions and Proposals

Conclusions

To begin with, all types of civil infrastructure are materialized impacts of climate change due to several interacting physical processes. The compound disruption of transportation infrastructure by greater pluvial water infiltration is compounded by the demonstrated empirical evidence that capacity and degradation of the road network progressively degrades even at relatively shallow depths of flooding. The urban drainage systems that were planned based on the historical data of the precipitation are flowing into the sewerage, which is being systematically undersized with the projected future amount of precipitation. Concrete structures that are reinforced have accelerated deterioration with the rate of carbonation as well as enhanced chloride ingress under high temperatures and high levels of CO_2 in the atmosphere. Tunnel and underground infrastructure are exposed to new groundwater and thermal stressors. The stability of slope and embankment depends on the alteration of the regimes of precipitation and temperature.

Current strategies of adaptation, nature-based drainage systems, revised design requirements, greater monitoring, and better governance structures are technologically feasible and cost-efficient. The difference between the technologically feasible and the really happening adaptation is more due to a contest of governance and funds than a knowledge contest.

The thesis stated was then aimed at answering the following practical question:

Is the civil infrastructure properly prepared for the climate conditions projected to occur within the next 50 years, and in the event it does, what is to be done about it?

The response to part one is obviously no, and the content of the thesis is the answer to the second part. The author of the thesis concludes the following while answering the question:

1. The Baltic region is warming at a faster rate than the EU. The average annual temperature variation in Latvia has been on an upward trend since 1961, increasing at an average rate of 1.4 ˚C/year, and the temperature change is projected to continue in the future as well, depending on the rate of emissions. Extreme rain has been on the increase, and the increase has been about 20 %. Sea level is rising, and coastal erosion is taking place. They are not projections as they are perceived facts recorded in the Latvian Environmental, Geology, and Meteorology Centre (LVGMC) data.
2. The unique and counter-intuitive conclusion of this thesis is that although the mean increasing depth of frost with warmer winters is decreasing, the number of freeze words

followed by thaws and verses per winter season has increased by the percentage rate of 46 % since 1961 till 1990. This cycling is more destructive to the pavement structure than deep stable frost, and translates into a small 34 % deduction on the effective pavement service life, and an incremental annual maintenance liability of about € 4.3 million. There are direct implications of this finding on LBN 228-06 norm revision.

3. The analysis of interdependency within infrastructure identified a few critical nodes of the network - especially the Riga central drainage system and the crossing in the region of A1 and Daugava - due to which the failure of infrastructure under the influence of climate factors was established. The process of adaptation of these nodes has a multiplier effect in terms of the overall system resilience.
4. Latvia Infrastructure, the total estimated annual infrastructure damage in Latvia due to climatic factors is close to € 46 million, increasing at an approximate rate of 28 % over the 2000-2022 period. Latvia also has higher scores on all the vulnerability indicators assessed, in comparison to the EU-27 average scores. The thesis hypothesizes that the existing norms and practices are inadequate to support the estimated climate trajectory.
5. The review of the international best practice shows that a holistic approach to infrastructure climate adaptation is totally possible. Each of the Netherlands, Finland, and the UK demonstrates that the answer lies in mainstreaming climate requirements into current design, procurement, and regulatory practices - not in the development of parallel systems. Latvia can look to Finland in particular, who have already revised norms regarding the design of its pavement and has recently launched a national programme on upgrading its culvert design, at a manageably low cost.
6. Climatically resilient materials such as polymer-modified bitumen in pavements, decreased water-cement ratio, and supplementary cementitious material (SCM)-enhanced concrete in structures are commercially available and lifecycle cost-effective today. They need not only to be an environmentally sound decision but also a logically sound economic decision to embrace them in state procurement specifications.
7. The three case studies substantiate the fact that the investment in adaptation is economically justified. The cost-benefit ratio obtained from the expenses incurred was calculated to be between 2.2:1 and 4.1:1. The identified total priority adaptation investment amounts to between € 55 million in 10 years, with a net contribution in the national budget of between € 8-14 million after co-financing by the EU. Part of this financing could be through the provision of a cost-effective mechanism by a part of the second issue of the Latvian sovereign green bond.

8. Monitoring matters. There is no way that adaptation investment without performance tracking can be improved or defended to funders and to the public. The KPI framework offers one such practical starting point in the IS based on the monitoring of adaptation of the infrastructure of a country, which would be based on the ISO 14090 guidelines.

Proposals

According to the carried-out research, the following proposals are to be made to the respective Latvian organizations:

1. Ministry of Economics
    a. Switch to formal revision of LBN 223-99 (and LBN 223-15) and LBN 228-06 in 24 months, which incorporates the intensities of rainfall adjusted for climate change for the 2050 and 2080 design horizons (RCP 4.5 as base case scenario and RCP 8.5 as sensitivity), and a freeze-thaw cycle frequency correction factor.
    b. Amend the Latvia Building Norms (LBN) so that drainage and pavement design standards incorporate climate projections, to address the climate change-related capacity gaps identified in the Riga urban drainage case study.
    c. Expand the use of quantitative climate risk assessment methods (IPCC – hazard, exposure, and vulnerability) across all new and updated technical norms of civil infrastructure.
2. *Latvijas Valsts Ceļi* (Latvian State Roads)
    a. Include "Polymer-Modified Bitumen" (PMB 25/55-60 or equivalent) as a standard in pavement procurement specifications for all roads where AADT is greater than 2,000, except for those meant to be asphaltic concrete in cold climates, to improve performance under higher temperatures and reduce rutting/brittleness risk.
    b. Develop and enforce a bridge joint inspection/replacement programme based on risk compared to the other pre-2000 and pre-1970 bridge cohorts, full assessments of thermal expansion joint capacity of the network by 2028 under assumed warming scenarios of RCP 4.5 and 8.5.
    c. Develop and upscale the National Bridge Scour Monitoring (BSC) network using real-time IoT sensors installed at high-risk bridges (bigger than a catchment area of 100 km^2); connect the scour risk monitoring network with LVGMC hydrological data to incorporate climate-corrected scour risk in routine maintenance practices of bridges.

3. Ministry of Smart Administration and Regional Development (working with Latvian Environmental, Geology, and Meteorology Center - LVGMC)
    a. Provide and periodically update a national-scale sea-level rise and coastal erosion mapping (at 1:5,000) based on LiDAR surveys to support spatial planning and infrastructure planning along the coasts of the Gulf of Riga and elsewhere.
    b. Prepare and adopt a national coastal erosion management strategy that includes beach nourishment initially and, as a long-term preferred adaptation action, for vulnerable coastal roads and assets, the strategy must prioritise managed realignment.
    c. Include mandatory nature-based solutions (NBS) / Blue-Green Infrastructure requirements (at least 15% of green infrastructure in new developments) in urban planning standards, especially for actively flooded areas of situational shortcomings like Riga, Jelgava, and Jūrmala.
4. Cabinet of Ministers / Cross-ministerial
    a. Realise EU Taxonomy and Cohesion Fund requirement for mandatory climate proofing of all State-funded infrastructure projects with design life over 25 years by 2026, based on ISO 14090.
    b. Provide dedicated climate rescue funds for urban drainage, especially in major cities (at least €32 million) and for total climate-suitable infrastructure investment (total ~ €53 million window) in the form of a second sovereign green bond.
    c. Implement the suggested Monitoring & Evaluation KPI plan (compliant to ISO 14090), highlighting decision triggers and escalation plans, reporting the performance every year to the European Commission regarding climate adaptation.

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